sPHENIX Internal Note

Photon Efficiency and Energy Response with ΔR Truth-Matching Removed: Single, Double, and Mixed Interaction MC

Isolated Photon Cross-Section in p+p at $\sqrt{s} = 200$ GeV

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Abstract

Building on the companion study [1] that demonstrated the $\Delta R < 0.1$ truth-matching cut rejects $\sim$3.5% of correctly track-ID-matched truth photons in single-interaction MC, this note quantifies the efficiency impact of removing the cut across three MC sample categories: single-interaction, double-interaction, and physics-mixed (0 mrad, 77.6%/22.4%). The factorized efficiency chain $\varepsilon_{\rm reco} \to \varepsilon_{\rm iso|reco} \to \varepsilon_{\rm id|reco+iso} \to \varepsilon_{\rm all}$ is computed with and without the ΔR requirement in a single pass over `photon10` samples. Removing the ΔR cut improves the total efficiency $\varepsilon_{\rm all}$ by +2.1% (single), +13.8% (mixed), and approximately doubles it for double-interaction MC. The reconstruction stage dominates the change; isolation efficiency is unaffected; photon ID (BDT) efficiency shows a measurable degradation in double-interaction MC from vertex-shifted cluster kinematics. The single/double/mixed overlay provides a clean decomposition of pileup effects on each efficiency stage.

We then extend the study to the photon *energy response* $R_{E_{\rm T}} \equiv E_{\rm T}^{\rm reco}/p_{\rm T}^{\rm truth}$ across six sample configurations: single, double, and four mixed variants covering the two RHIC crossing-angle running periods (0 mrad with 22.4% cluster-weighted double-interaction fraction and 1.5 mrad with 7.9%), each with and without the data-driven z-vertex reweighting used in the main pipeline. The response is examined at three selection levels: *reco* (trkID match only), *common cut* (reco + common cuts, i.e. the working point that enters ABCD), and *tight iso* (common cut + isolation + tight BDT, i.e. the signal region). At the reco level, the pileup-induced vertex shift produces a direct kinematic bias ($\sim$3 pp mean shift and $\sim$30% relative width increase comparing double to single); after the common cuts the mean bias is largely absorbed by the cut boundaries, leaving a residual $\sim$0.3 pp mean shift and $\sim$22% relative width increase. The 1.5 mrad mixed sample (the nominal production period) tracks single-interaction MC to within $\sim$0.1 pp in mean at the common cut and tight iso levels. The 0 mrad mixed sample exhibits a measurably different response that motivates treating the two running periods separately in the pileup systematic. Vertex reweighting modifies the response by less than $\sim$0.5 pp at the common cut level, confirming that it is sub-dominant relative to the interaction-type composition itself.

Contents

1 Introduction

The reconstruction efficiency for the sPHENIX isolated photon cross-section measurement is computed from PYTHIA + GEANT4 MC using a factorized chain of selection stages: reconstruction, isolation, and photon identification (BDT). The truth association between generated photons and reconstructed EMCal clusters is established by the GEANT track ID (`trkID`), optionally supplemented by a geometric proximity requirement $\Delta R < 0.1$.

A companion study [1] demonstrated that this ΔR cut is a vertex-resolution artifact: in single-interaction MC it rejects $\sim$3.5% of `trkID`-matched photons, and in double-interaction MC the failure rate reaches 65%, driven entirely by the vertex displacement $|\Delta z| > 9.35$ cm (the geometric critical threshold set by the EMCal barrel radius). The rejected clusters are well-reconstructed photons with $E_\mathrm{T}/p_\mathrm{T}^\mathrm{truth} \in [0.8, 1.2]$ for 73% of cases.

Based on those findings, the $\Delta R < 0.1$ requirement was removed from the production efficiency calculation. This note quantifies the resulting efficiency changes across all interaction types and examines the interplay between the ΔR removal and the photon ID (BDT) selection, which uses an E_T-dependent threshold susceptible to vertex-induced kinematic shifts.

With the ΔR artifact out of the way, Section 6 extends the study to the photon *energy response* $R_{E_\mathrm{T}} = E_\mathrm{T}^\mathrm{reco}/p_\mathrm{T}^\mathrm{truth}$. The response directly drives the cluster E_T used in the ABCD region classification, the E_T-dependent BDT threshold, and the unfolding response matrix, so any pileup-induced bias in R_{E_T} feeds straight into the cross-section. Six sample configurations are studied (single, double, and four mixed variants covering the 0 mrad / 1.5 mrad running periods with and without data-driven z-vertex reweighting), at three selection levels that mirror the factorized efficiency chain. This allows us to disentangle interaction-type composition from residual data–MC vertex mismatch — the two independent knobs available to the pileup systematic.

2 Samples and Method

2.1 Monte Carlo Samples

Three sample configurations are studied, all using the `photon10` generator (truth p_T range 14–30 GeV):

- **Single-interaction:** `photon10` nominal MC ($\sim$10M events). One *pp* collision per bunch crossing.
- **Double-interaction:** `photon10_double` MC ($\sim$10M events). Full GEANT simulation of two overlapping *pp* collisions per bunch crossing, with the reconstructed vertex at the average of the two collision vertices.
- **Mixed (0 mrad):** Physics-weighted combination of 77.6% single + 22.4% double, corresponding to the cluster-weighted double-interaction fraction computed by `calc_pileup_range.C` for the 0 mrad crossing-angle running period at RHIC.

2.2 Efficiency Definitions

The efficiency is factorized into four stages, each conditional on the previous:

$$\varepsilon_{\text{reco}} = \frac{N(\text{reco} + \text{common cuts})}{N(\text{truth photons})} \tag{1}$$

$$\varepsilon_{\text{iso}|\text{reco}} = \frac{N(\text{reco} + \text{common} + \text{isolated})}{N(\text{reco} + \text{common})} \tag{2}$$

$$\varepsilon_{\text{id}|\text{reco+iso}} = \frac{N(\text{reco} + \text{common} + \text{iso} + \text{tight})}{N(\text{reco} + \text{common} + \text{iso})} \tag{3}$$

$$\varepsilon_{\text{all}} = \varepsilon_{\text{reco}} \times \varepsilon_{\text{iso}|\text{reco}} \times \varepsilon_{\text{id}|\text{reco+iso}} = \frac{N(\text{all cuts})}{N(\text{truth photons})} \tag{4}$$

The "common cuts" include cluster probability, shower-shape quality gates (e11/e33, ω_r, ω_η), and optionally the NPB score cut. "Isolated" applies the parametric topo-cluster isolation cut $E_{\text{T}}^{\text{iso}} < b + s \cdot E_{\text{T}}$ (topo $R_{\text{cone}} = 0.4$). "Tight" applies the full shower-shape selection plus the E_{T}-dependent BDT threshold $\text{BDT}_{\text{min}} = \text{intercept} + \text{slope} \times E_{\text{T}}$.

2.3 Matching Modes

Each truth photon is evaluated under two matching modes simultaneously in a single pass:

`withDR`: Requires both `trkID` match *and* $\Delta R < 0.1$ (the legacy requirement).

`noDR`: Requires only the `trkID` match (the new baseline).

Clusters that enter the `noDR` numerator but not the `withDR` numerator are labeled "new clusters"—these are the photons recovered by removing the geometric cut. Their BDT pass rate is tracked separately.

The analysis macro is `compare_efficiency_deltaR.C`, which writes all `TEfficiency` objects to a single output ROOT file.

3 Results: Per-Stage Efficiency Comparison

Table 1 presents the p_{T}-integrated efficiency at each factorized stage for all three sample sets, computed with and without the $\Delta R < 0.1$ requirement.

Table 1: Integrated efficiency at each factorized stage for single-interaction, double-interaction, and mixed (0 mrad) MC. "withDR" denotes the legacy matching; "noDR" denotes `trkID`-only matching.

	Single		**Double**		**Mixed 0 mrad**	
Stage	withDR	noDR	withDR	noDR	withDR	noDR
$\varepsilon_{\text{reco}}$	91.4%	93.5%	30.7%	73.1%	73.4%	87.5%
$\varepsilon_{\text{iso}\|\text{reco}}$	74.2%	74.2%	71.1%	71.0%	73.8%	73.4%
$\varepsilon_{\text{id}\|\text{reco+iso}}$	82.9%	82.7%	82.6%	69.3%	82.9%	79.5%
ε_{all}	56.2%	57.4%	18.0%	36.0%	44.9%	51.1%

Table 2 shows the ratio of `noDR` to `withDR` efficiency at each stage, quantifying the impact of removing the cut.

The key observations are:

Table 2: Ratio of efficiency without the ΔR cut to efficiency with the cut ($\varepsilon_{\text{noDR}}/\varepsilon_{\text{withDR}}$) at each stage. Values above 1.0 indicate improvement from cut removal; values below 1.0 indicate degradation in the recovered cluster population.

Stage	Single	Double	Mixed 0 mrad
$\varepsilon_{\text{reco}}$	1.023	2.386	1.191
$\varepsilon_{\text{iso\|reco}}$	1.000	0.998	0.994
$\varepsilon_{\text{id\|reco+iso}}$	0.997	0.840	0.960
ε_{all}	1.021	1.999	1.137

- **Reconstruction** ($\varepsilon_{\text{reco}}$): This stage captures the full impact of the ΔR removal. Single MC improves by 2.3%, reflecting the small population of vertex-displaced clusters. Double MC improves by a factor of ∼2.4, recovering the 65% of clusters that failed the ΔR cut due to vertex displacement. The residual 27% gap between double and single noDR efficiency reflects genuine missing clusters (no `trkID` match at all: 2.4% for single, 7.9% for double).
- **Isolation** ($\varepsilon_{\text{iso|reco}}$): Essentially unchanged across all samples (ratios within 0.5% of unity). The isolation requirement tests calorimeter energy in an annular cone around the cluster; vertex displacement does not significantly alter the energy sum, and the extra energy from the second interaction causes only a ∼3% absolute degradation comparing single to double.
- **Photon ID** ($\varepsilon_{\text{id|reco+iso}}$): Single MC shows negligible change (0.2% decrease). Double MC shows a significant 16% decrease (ratio 0.840), indicating that the recovered clusters have measurably worse BDT pass rates. Mixed MC interpolates at 4.0% decrease.
- **Total** (ε_{all}): Net improvement in all cases. The reconstruction gain dominates the BDT degradation. Single: +2.1%; mixed: +13.8%; double: nearly doubling from 18.0% to 36.0%.

Figure 1 shows the p_{T}-dependent efficiency ratio for all four stages, for single and mixed 0 mrad MC. The pure double-interaction sample is omitted from this ratio view because its ΔR rejection rate (65%) produces a ratio far off the displayed scale.

4 Overlay: Efficiency without ΔR across Interaction Types

With the ΔR cut removed, the remaining efficiency differences between single, double, and mixed MC arise from genuine pileup physics effects rather than truth-matching artifacts. Figure 2 presents the four-panel overlay of the `noDR` efficiency for all three sample types, providing a clean decomposition of pileup degradation at each selection stage.

4.1 Reconstruction Efficiency

Single-interaction MC achieves $\varepsilon_{\text{reco}} = 93.5\%$; double-interaction drops to 73.1%, a 20 percentage point (pp) reduction. The p_{T} dependence is weak: single MC is flat near 93–95% across the full range, while double MC spans 70–77%, maintaining a roughly constant ∼20 pp gap. This gap reflects two effects:

1. **Genuine cluster loss:** In double-interaction events, the additional hadronic activity can merge the signal photon cluster with nearby energy deposits, causing the `trkID` to be lost

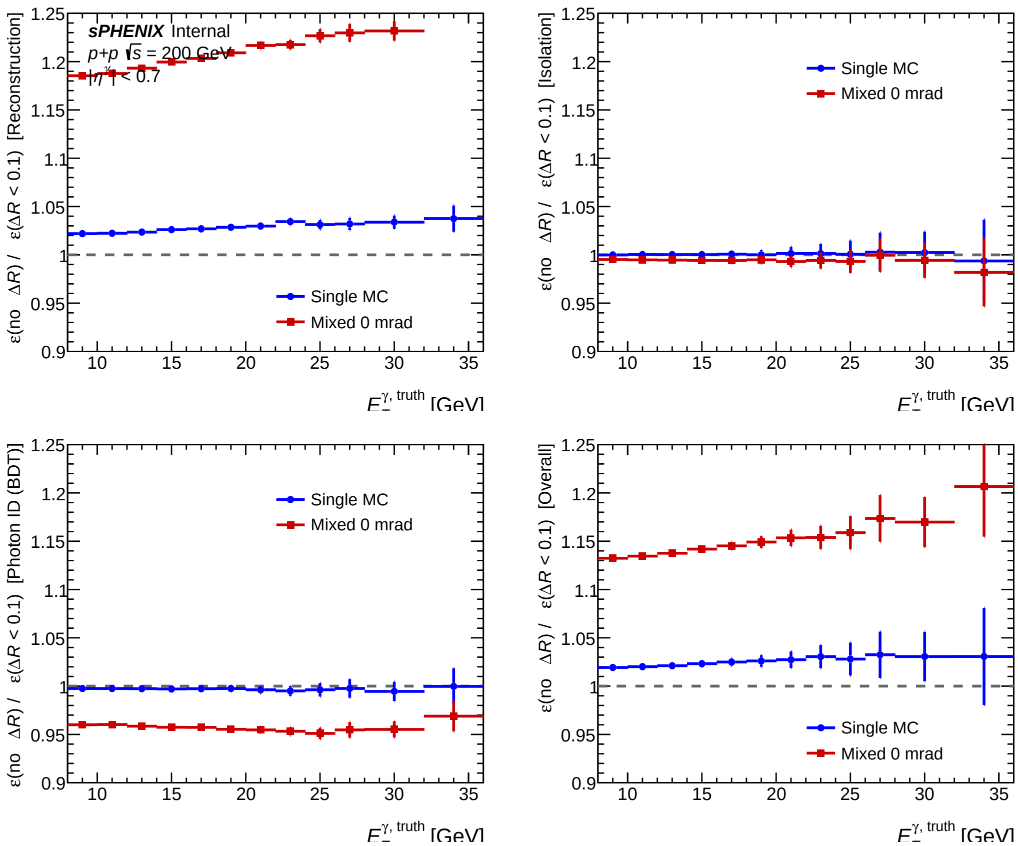

Figure 1: Ratio of efficiency without ΔR cut to efficiency with ΔR cut as a function of truth p_{T} for single (blue) and mixed 0 mrad (red) MC. The reconstruction stage shows the largest change, particularly for the mixed sample containing 22.4% double-interaction events. Isolation is flat. Photon ID shows a deficit in the mixed sample at all p_{T}.

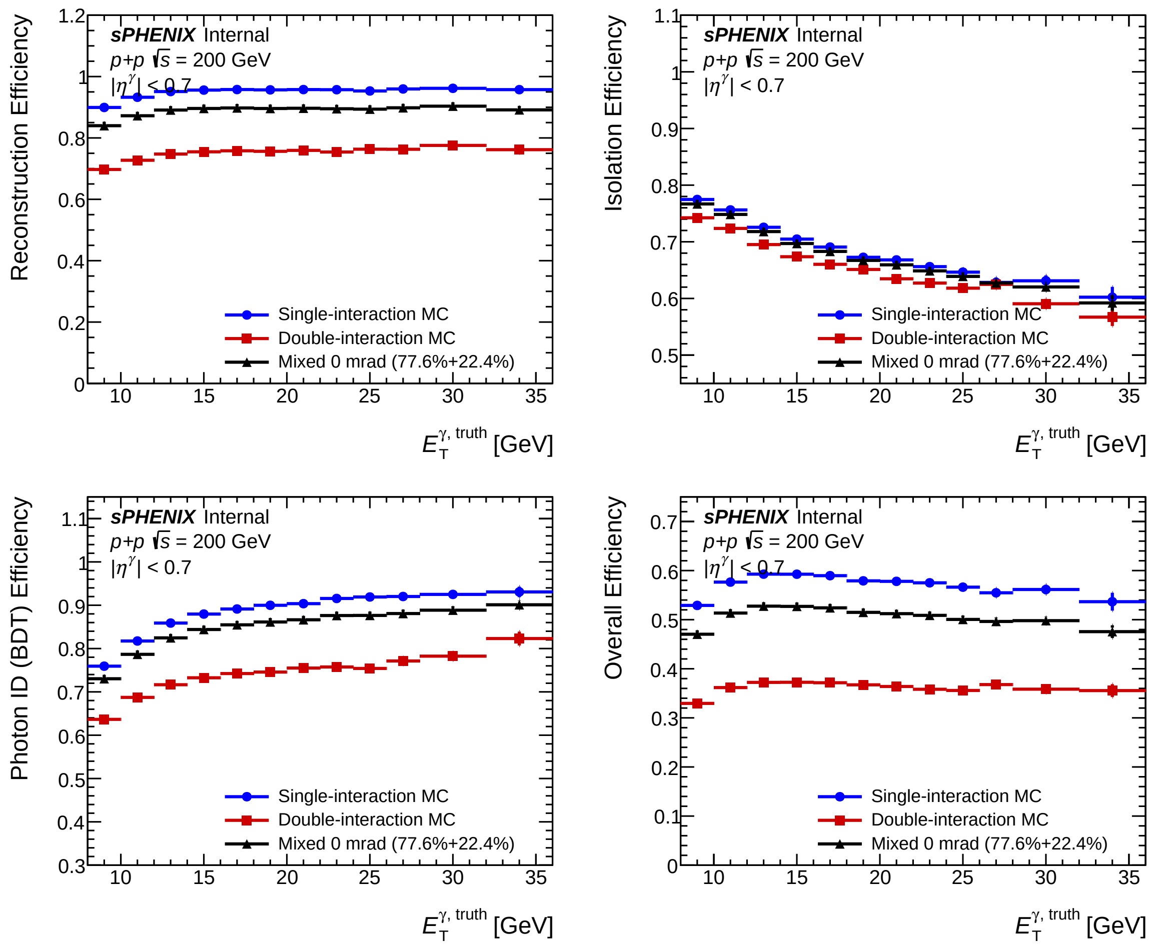

Figure 2: Per-stage efficiency as a function of truth p_{T} without the ΔR cut, overlaying single-interaction (blue), double-interaction (red), and mixed 0 mrad (black) MC. From upper left to lower right: reconstruction, isolation, photon ID (BDT), and total efficiency.

from the leading cluster. The no-`trkID`-match rate is 7.9% for double MC vs. 2.4% for single MC, accounting for ∼5.5 pp of the gap.

2. **Common cut failures from shifted kinematics:** The remaining ∼15 pp gap arises because the reconstructed vertex in double-interaction events shifts the cluster η and E_T, causing some clusters to fail the common quality cuts (e.g., the shower-shape bounds applied at the reconstructed kinematics).

The mixed 0 mrad sample tracks close to single MC at 84–89%, consistent with the weighted average $0.776 \times 93.5\% + 0.224 \times 73.1\% = 89.0\%$ (the small offset from 87.5% reflects bin-by-bin weighting effects).

4.2 Isolation Efficiency

Isolation efficiency is essentially flat across interaction types: single 74.2%, double 71.0%, mixed 73.4%. All three samples fall together from ∼77% at low p_T (∼9 GeV) to ∼60% at high p_T (∼34 GeV), with the p_T-falling trend dominating over the pileup effect. Double MC sits only 3–5 pp below single across the range. The small degradation reflects the additional hadronic energy deposited in the isolation cone by the second interaction. This is a genuine, small pileup effect that does not interact with the ΔR matching.

4.3 Photon ID (BDT) Efficiency

This is the most physically interesting stage. Single MC achieves $\varepsilon_{\mathrm{id|reco+iso}} = 82.7\%$; double MC drops to 69.3%, a significant 13 pp reduction. The mechanism is the E_T-dependent BDT threshold:

$$\mathrm{BDT}_{\mathrm{min}}(E_T) = \mathrm{intercept} + \mathrm{slope} \times E_T \tag{5}$$

In double-interaction events, the vertex shift modifies the reconstructed $E_T = E/\cosh(\eta_{\mathrm{reco}})$. Because η_{reco} is computed from the displaced vertex, the cluster evaluates the BDT threshold at a shifted E_T, changing the working point. The shifted shower-shape variables (particularly ω_η, which is sensitive to η) further degrade the BDT score itself.

The p_T dependence reveals an interesting pattern: at low p_T (∼9 GeV) the gap is largest, with double MC at ∼64% vs. single at ∼76% (12 pp). At high p_T (∼34 GeV), the gap narrows to ∼10 pp (double ∼83% vs. single ∼93%). This p_T dependence is consistent with the parametric BDT threshold of Eq. (5): at low E_T, the threshold is flatter and a given E_T shift has a larger relative impact on the pass/fail decision.

The mixed 0 mrad sample shows $\varepsilon_{\mathrm{id|reco+iso}} = 79.5\%$, a 3.2 pp reduction relative to single MC, sitting closer to the single curve due to the 77.6% single fraction dominating.

4.4 Total Efficiency

The total efficiency without ΔR is: single 57.4%, double 36.0%, mixed 51.1%. The single-to-double drop is dominated by the reconstruction (20 pp) and BDT (13 pp) stages, with isolation contributing only 3 pp. The p_T dependence is moderate: single spans 53–58%, double 33–37%, and mixed 47–51%. Notably, the ratio $\varepsilon_{\mathrm{all}}^{\mathrm{double}}/\varepsilon_{\mathrm{all}}^{\mathrm{single}} \approx 0.63$ is remarkably flat vs. p_T, indicating that the pileup degradation is approximately factorizable as a constant scale factor.

The mixed 0 mrad efficiency of 51.1% is the physics-relevant number for the pileup systematic evaluation.

5 Photon ID Deep Dive: New Clusters

The "new clusters" recovered by removing the ΔR cut provide a direct window into how vertex displacement affects BDT classification. Table 3 compares the BDT pass rate for new clusters only versus all clusters in the `noDR` matching.

Table 3: Photon ID (BDT) pass rate for "new" clusters (recovered by removing $\Delta R < 0.1$) compared to all `noDR`-matched clusters. The new clusters have 10–13 pp lower BDT pass rate, driven by vertex-shifted kinematics.

Sample	$\varepsilon_{\mathrm{id}\vert\mathrm{reco+iso}}$ (new clusters only)	$\varepsilon_{\mathrm{id}\vert\mathrm{reco+iso}}$ (all, noDR)
Single	73.9%	82.7%
Double	59.8%	69.3%
Mixed	61.3%	79.5%

Several observations emerge:

- The new clusters consistently show a lower BDT pass rate than the full population: 9 pp lower for single MC, 10 pp for double, and 18 pp for the mixed sample. This confirms that vertex displacement degrades the BDT score via the mechanism described in Section 4.
- In single MC, only ∼2.3% of matched clusters are "new" (those with $\Delta R > 0.1$), so their lower BDT pass rate has negligible impact on the integrated $\varepsilon_{\mathrm{id}|\mathrm{reco+iso}}$ (change of −0.2 pp).
- In double MC, ∼58% of matched clusters are "new" (reflecting the 65% ΔR failure rate from the companion study, reduced by those with no `trkID` match at all). The large new-cluster fraction drives the 13 pp $\varepsilon_{\mathrm{id}|\mathrm{reco+iso}}$ degradation.
- The mixed sample's new-cluster BDT pass rate (61.3%) is dominated by the double-interaction contribution, since the single-interaction MC contributes very few new clusters. Visually, the double and mixed curves overlap almost completely in Figure 3.
- The p_{T} dependence is strongly rising: single new clusters span ∼68% at low p_{T} to ∼93% at high p_{T}; double new clusters span ∼55% to ∼78%. Higher-p_{T} photons are more robust to vertex-induced kinematic shifts because their BDT scores sit further above the threshold.

Figure 3 overlays the BDT pass rate for new clusters across the three sample types as a function of truth p_{T}.

6 Photon Energy Response Across Interaction Categories

Removing the ΔR cut is intentional: the matching criterion was biasing the efficiency in double-interaction MC. Having removed that bias, we can now ask the complementary question of how the pileup vertex shift affects the reconstructed energy itself. This section extends the study to the *photon energy response*, defined as

$$R_{E_{\mathrm{T}}} \equiv \frac{E_{\mathrm{T}}^{\mathrm{reco}}}{p_{\mathrm{T}}^{\mathrm{truth}}} \tag{6}$$

for every truth photon that is `trkID`-matched to a reconstructed EMCal cluster with $E_{\mathrm{T}} >$ reco_min_ET. (For photons, $p_{\mathrm{T}}^{\mathrm{truth}} \equiv E_{\mathrm{T}}^{\mathrm{truth}}$; the reco-level quantity uses $E_{\mathrm{T}}^{\mathrm{reco}} = E^{\mathrm{reco}}/\cosh(\eta^{\mathrm{reco}})$,

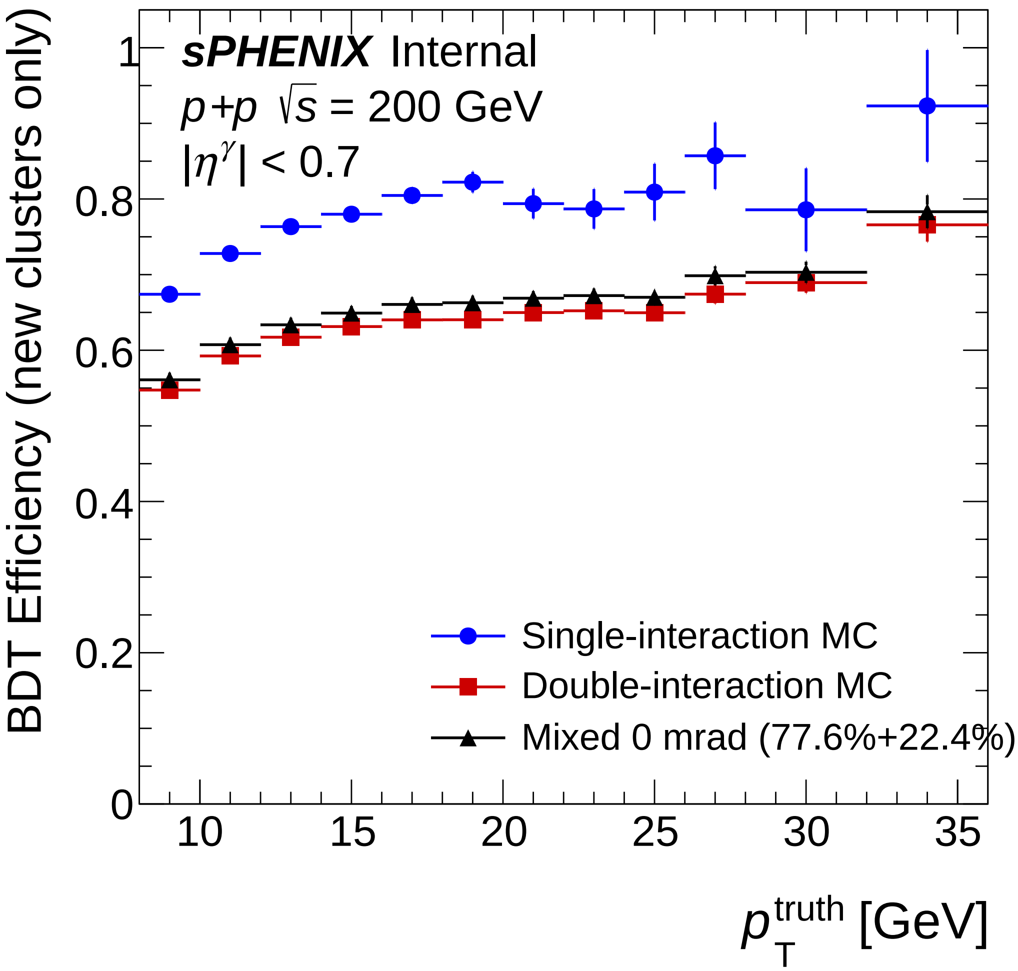

Figure 3: BDT pass rate for "new" clusters (those with $\Delta R > 0.1$ but valid `trkID` match) as a function of truth p_{T} for single (blue), double (red), and mixed (black) MC. These clusters have systematically lower BDT pass rates due to vertex-shifted kinematics. The deficit narrows at higher p_{T} where the absolute BDT pass rate is higher.

where $\eta^{\rm reco}$ is computed from the *reconstructed* vertex. We also track $R_E \equiv E^{\rm reco}/E^{\rm truth}$ — the pure calorimeter response, with $E^{\rm truth} = p_{\rm T}^{\rm truth}\cosh(\eta^{\rm truth})$ — but emphasize $R_{E_{\rm T}}$ in the main text because it is the quantity that enters the $E_{\rm T}$-dependent BDT threshold and the $p_{\rm T}$ bins of the cross-section.)

6.1 Sample Sets and Method

Six sample sets are studied to disentangle the two physics effects at play (interaction-type composition, and z-vertex distribution differences between data and MC):

- **single:** `photon10` nominal, weight 1.
- **double:** `photon10_double`, weight 1.
- **mixed 0 mrad:** $0.776\times$ single $+\ 0.224\times$ double, corresponding to the cluster-weighted double-interaction fraction of 22.4% at 0 mrad crossing angle (from `calc_pileup_range.C`).
- **mixed 0 mrad (vtx rw):** same composition, with the z-vertex distribution reweighted to match the 0 mrad data.
- **mixed 1.5 mrad:** $0.921\times$single$+0.079\times$double, corresponding to the 7.9% cluster-weighted double-interaction fraction at 1.5 mrad (the nominal production running period).
- **mixed 1.5 mrad (vtx rw):** same, reweighted to 1.5 mrad data vertex distribution.

The analysis macro is `compare_energy_response.C`; it shares the selection code path with `compare_efficiency_deltaR.C` (Section 2) and writes `energy_response_comparison.root`. The response histograms are filled at three selection levels, each mirroring a stage of the factorized efficiency chain:

reco: `trkID` match only — no ΔR cut and no common cuts. This isolates the raw detector response as seen by the truth association.

common cut: reco + common cuts (cluster probability, e_{11}/e_{33}, ω_r, ω_η bounds, NPB if enabled). This is the working point that enters the ABCD region counting.

tight iso: common cut + isolation + full tight selection (including the $E_{\rm T}$-dependent BDT threshold, Eq. 5). This is the final analysis selection used for the cross-section.

6.1.1 Vertex reweighting

The reweighting follows the two-pass recipe used in the main pipeline (`ShowerShapeCheck.C`). In Pass 1, the reconstructed vertex distribution $h_z^{\rm single}(z)$ and $h_z^{\rm double}(z)$ are filled from `photon10` and `photon10_double` respectively, without any selection beyond the nominal event-level vertex cut $|z_{\rm vtx}| < 60$ cm. For each mixed set with reweighting enabled, the combined sim distribution is constructed as

$$h_z^{{\rm sim},X}(z) \;=\; \alpha_{\rm nom}^X\, h_z^{\rm single}(z) \;+\; \alpha_{\rm double}^X\, h_z^{\rm double}(z) \tag{7}$$

with $X \in \{0\text{ mrad}, 1.5\text{ mrad}\}$ and $(\alpha_{\rm nom}^X, \alpha_{\rm double}^X) \in \{(0.776, 0.224), (0.921, 0.079)\}$. The per-event reweighting factor is

$$w_{\rm vtx}^X(z) \;=\; \frac{h_z^{{\rm data},X}(z)/\sum_{z'} h_z^{{\rm data},X}(z')}{h_z^{{\rm sim},X}(z)/\sum_{z'} h_z^{{\rm sim},X}(z')} \tag{8}$$

where $h_z^{\mathrm{data},X}$ is the Pass-1 data vertex distribution from `data_histo_bdt_0rad_vtxscan.root` and `data_histo_bdt_1p5rad_vtxscan.root` for the two crossing-angle periods. Both sim and data histograms are rebinned by a factor 5 (bin width 5 cm) and the ratio is smoothed with one pass of `TH1::Smooth` to suppress residual statistical fluctuations. Figure 4 shows the raw sim vertex distributions for the two input samples and the resulting $w_{\mathrm{vtx}}^X(z)$ ratios for the two periods.

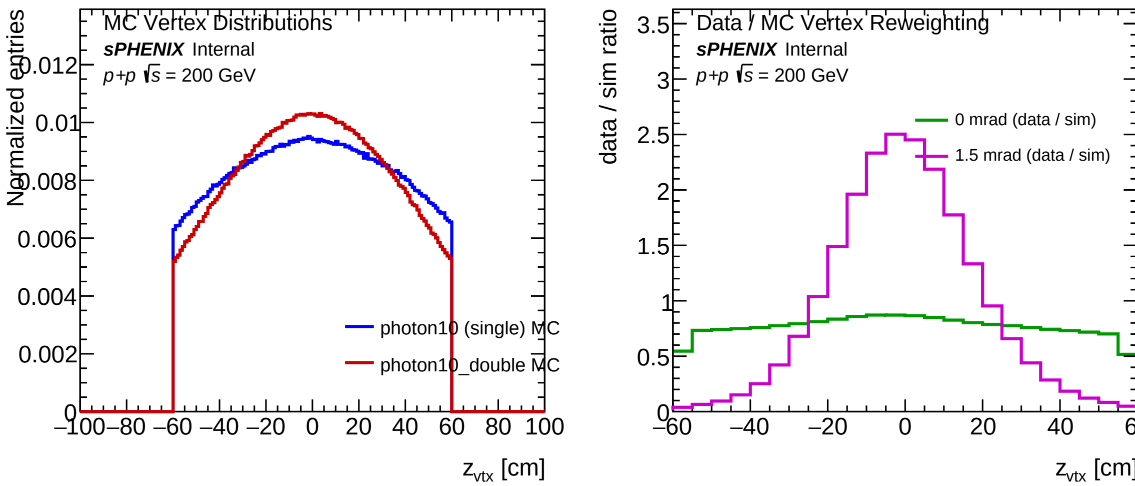

Figure 4: Left: reconstructed z-vertex distributions of `photon10` (blue, narrow, $\sigma_z \approx 7.6$ cm) and `photon10_double` (red, broad, $\sigma_z \approx 34.5$ cm). Right: data/sim reweighting ratios $w_{\mathrm{vtx}}^X(z)$ for the 0 mrad (green) and 1.5 mrad (magenta) mixed sets.

6.2 Integrated Response Summary

Table 4 summarises the pT-integrated mean and RMS of R_{E_T} for all six sample sets at the three selection levels. Entries are photons (not events), aggregated over the full truth-p_T range used in the analysis. The RMS is that of the unbinned distribution; the Gaussian-fit widths shown in the differential plots below differ by $\lesssim 1\%$ in the bulk of the response.

Table 4: Integrated mean $\langle R_{E_\mathrm{T}} \rangle$ and RMS $\sigma(R_{E_\mathrm{T}})$ of the photon energy response, for the six sample sets at the three selection levels. "reco": `trkID` only. "common cut": + common cuts. "tight iso": + iso + tight BDT.

	reco		**common cut**		**tight iso**	
Set	$\langle R_{E_\mathrm{T}} \rangle$	σ	$\langle R_{E_\mathrm{T}} \rangle$	σ	$\langle R_{E_\mathrm{T}} \rangle$	σ
single	0.939	0.124	0.948	0.113	0.989	0.071
double	0.910	0.164	0.951	0.138	0.982	0.093
mixed 0 mrad	0.931	0.137	0.949	0.120	0.987	0.076
mixed 0 mrad (vtx rw)	0.931	0.137	0.950	0.120	0.987	0.076
mixed 1.5 mrad	0.936	0.129	0.948	0.115	0.988	0.073
mixed 1.5 mrad (vtx rw)	0.940	0.128	0.953	0.114	0.988	0.073

Figure 5 presents the differential mean and Gaussian width of R_{E_T} as functions of truth p_T, overlaying all six sets at both the reco and common-cut levels. The truncated-Gaussian fit used for $\langle R_{E_{\mathrm{T}}} \rangle$ and $\sigma(R_{E_{\mathrm{T}}})$ is performed on the [0.3, 1.7] window of $R_{E_{\mathrm{T}}}$, the same window used by `RecoEffCalculator_TTreeReader.C` in the main pipeline, to ensure robustness against non-Gaussian tails.

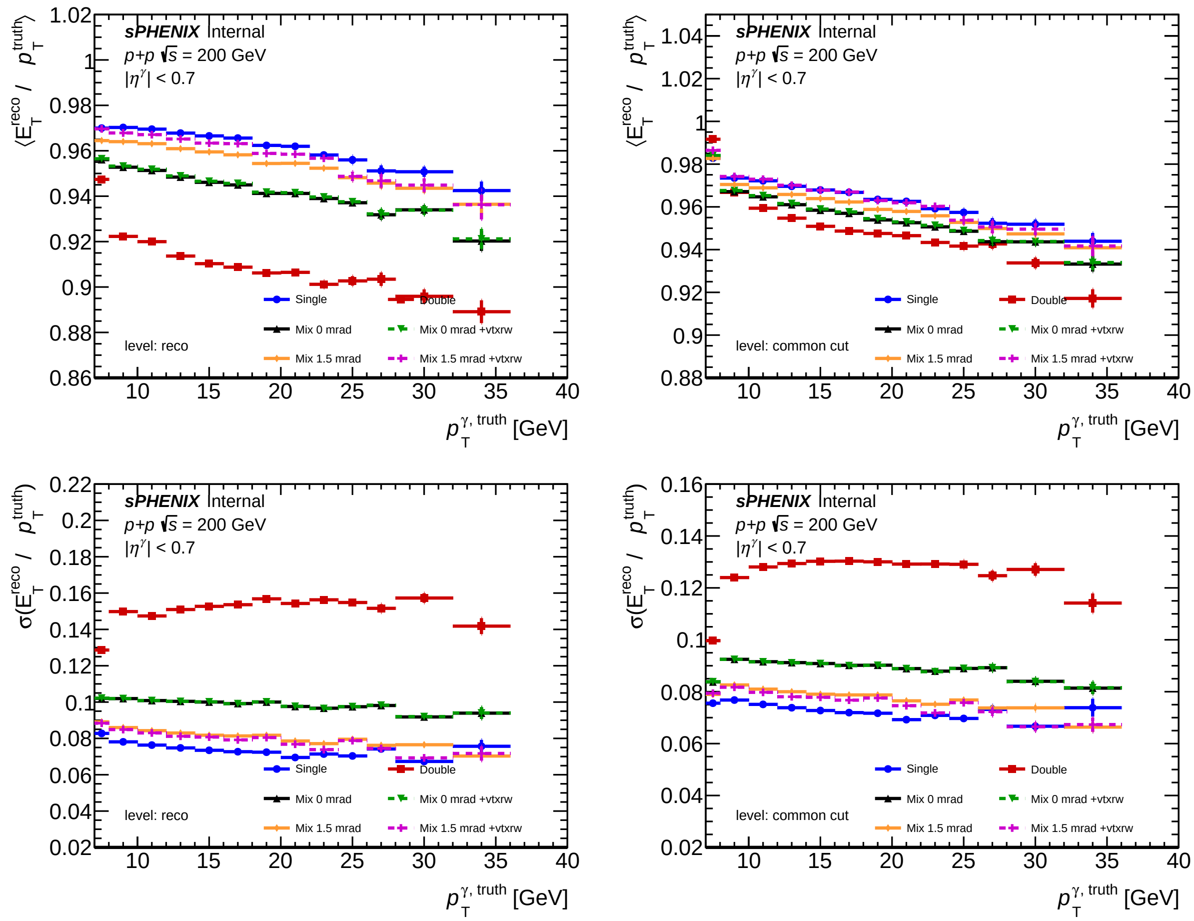

Figure 5: Mean (top row) and Gaussian width (bottom row) of the photon energy response $R_{E_{\mathrm{T}}} = E_{\mathrm{T}}^{\mathrm{reco}}/p_{\mathrm{T}}^{\mathrm{truth}}$ as a function of truth p_{T}, for all six sample sets. Left column: reco (`trkID` only); right column: common cut (+ common cuts). The reco-level panels expose the raw pileup bias; after the common cuts it is largely absorbed by the cut boundaries.

6.3 Pileup Effect: Single, Double, 0 mrad, 1.5 mrad

The comparison between single, double, mixed 0 mrad, and mixed 1.5 mrad (all without vertex reweighting) isolates the pileup impact as a function of interaction-type composition.

Reco level. The raw calorimeter response, evaluated on all `trkID`-matched clusters, shows the largest pileup bias: the integrated mean drops from 0.939 in single MC to 0.910 in double MC (a −2.9 pp shift, or 3.1% relative), and the RMS broadens from 0.124 to 0.164 (a 32% relative increase). The origin is kinematic: the reconstructed vertex in double-interaction events is the average of the two collision vertices, so the displacement $\Delta z = z_{\mathrm{vtx}}^{\mathrm{reco}} - z_{\mathrm{vtx}}^{\mathrm{true}}$ shifts the cluster pseudorapidity by

$$\Delta\eta \approx -\frac{\Delta z}{R_{\mathrm{EMCal}}\cosh(\eta^{\mathrm{truth}})}, \qquad R_{\mathrm{EMCal}} = 93.5 \text{ cm} \tag{9}$$

and hence shifts $E_{\mathrm{T}}^{\mathrm{reco}} = E^{\mathrm{reco}}/\cosh(\eta^{\mathrm{reco}})$ even though the deposited energy E^{reco} is unbiased. Because $1/\cosh$ is concave, the mean of $R_{E_{\mathrm{T}}}$ is systematically pulled below unity under the fluctuations in η^{reco}, consistent with Jensen's inequality applied to the mapping $\eta \mapsto 1/\cosh(\eta)$.

The mixed 0 mrad sample sits between the single and double extremes at $\langle R_{E_{\mathrm{T}}}\rangle = 0.931$ and $\sigma = 0.137$, consistent with the linear combination $0.776\times0.939+0.224\times0.910 = 0.932$ (mean) and a variance-weighted estimate $\sqrt{0.776 \times 0.124^2 + 0.224 \times 0.164^2 + 0.776 \times 0.224 \times (0.939 - 0.910)^2} = 0.136$ (RMS) — both agreeing with the simulated mixed set to within the fourth decimal.

The mixed 1.5 mrad sample is much closer to single: $\langle R_{E_{\mathrm{T}}}\rangle = 0.936$, $\sigma = 0.129$. With only 7.9% double-interaction content, the response is essentially that of single-interaction MC with $\sim$40% inflated tails. This is the physically relevant sample for the nominal production period.

Common-cut level. After the common cuts, the mean response converges across all samples to $\sim$0.948–0.951 (within $\pm$0.4 pp), indicating that the vertex-induced kinematic bias is almost entirely rejected by the shower-shape quality gates (e_{11}/e_{33}, ω_r, ω_η bound, cluster probability). The residual +0.3 pp excess of the double sample over single at the integrated level reflects a surviving tail of moderately-shifted clusters that passed the common cuts but still have biased $E_{\mathrm{T}}^{\mathrm{reco}}$. The sigma, however, remains measurably broader: 0.138 (double) vs 0.113 (single), a 22% relative increase. The common cuts shave off the mean-biasing tail faster than the symmetric-broadening bulk.

Figure 6 shows the $R_{E_{\mathrm{T}}}$ shape at $22 < p_{\mathrm{T}}^{\mathrm{truth}} < 24$ GeV (reco level) — the characteristic broadening of the double sample is directly visible as wider tails, especially to the low-$R_{E_{\mathrm{T}}}$ side (under-reconstruction from shifted η^{reco}).

Tight-iso level. Once the full tight selection (including the E_{T}-dependent BDT threshold) is applied, the integrated mean sits at 0.988–0.989 with $\sigma = 0.071$–0.093. The remaining inter-sample spread is 0.7 pp in mean and 22 pp in the relative sigma (double 0.093 vs single 0.071). The BDT preferentially selects clusters with vertex-consistent shower shapes, which correlates with small vertex shifts, so the surviving population is essentially unbiased in mean. Residual width differences are a sub-percent effect at the E_{T}-integrated level of the cross-section and are absorbed into the photon-energy-scale systematic.

p_{T} dependence. The p_{T}-dependent mean (top row of Fig. 5) shows a mild monotonic decrease with p_{T} in single MC (0.98$\rightarrow$0.91 from 8 to 40 GeV), reflecting the nominal EMCal energy-scale drift already measured in the main analysis. In double MC the same trend is present but offset downward by a roughly p_{T}-independent $\sim$3 pp; combined with the larger sigma, this offset propagates through the unfolding response matrix as a slightly different diagonal but qualitatively similar structure. The p_{T}-dependent sigma in double MC is flatter and broader than in single, consistent with a vertex-displacement contribution that does not scale with truth p_{T}.

6.4 Vertex-Reweighting Impact

The two vertex-reweighted mixed sets quantify how much of the residual data–MC vertex mismatch contributes to the response systematic. Table 5 shows the mean and RMS of $R_{E_{\mathrm{T}}}$ at the common-cut level for the two reweighting pairs, together with the differences.

The impact is small in both periods but not symmetric. For the 0 mrad sample the reweighting barely moves the integrated mean (+0.06 pp) and does not change the width appreciably — consistent with the 0 mrad data vertex distribution being close to the mixed-MC one after the 22.4% double-interaction blend. For the 1.5 mrad sample the reweighting has a larger effect (+0.45 pp in mean, −0.19 pp in sigma), reflecting the fact that the 1.5 mrad data vertex distribution is narrower

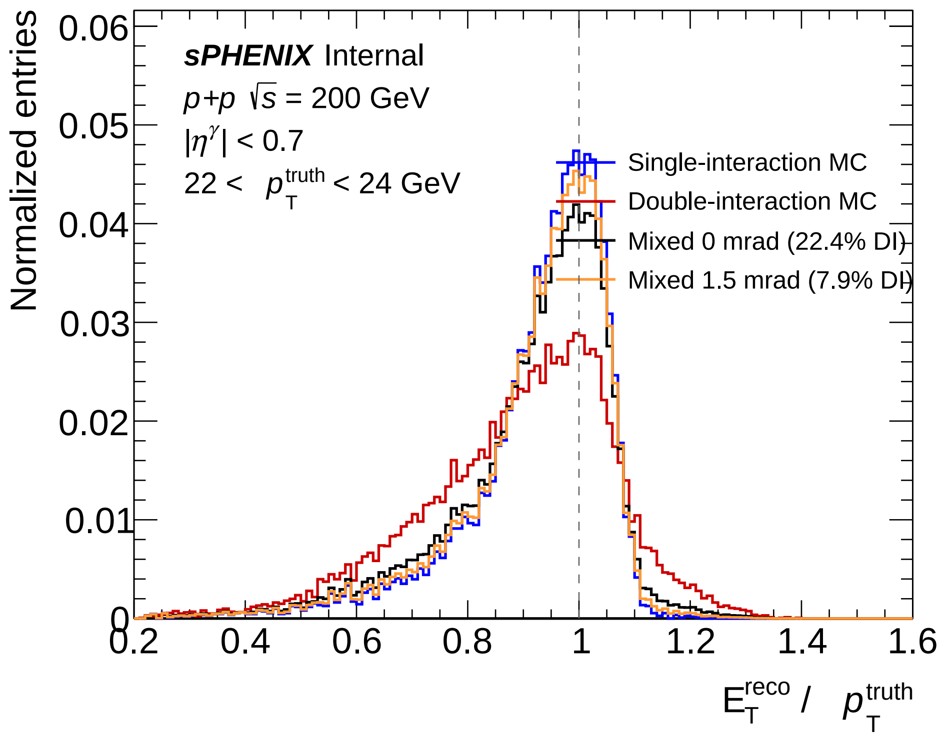

Figure 6: Normalized $R_{E_\mathrm{T}} = E_\mathrm{T}^\mathrm{reco}/p_\mathrm{T}^\mathrm{truth}$ distributions at reco level for the pileup comparison (single, double, mixed 0 mrad, mixed 1.5 mrad), in the bin $22 < p_\mathrm{T}^\mathrm{truth} < 24$ GeV. The vertical dashed line marks $R_{E_\mathrm{T}} = 1$. Double MC exhibits a markedly broader distribution with a low-side tail from vertex-induced η^reco bias; mixed samples interpolate by their double-interaction content.

Table 5: Integrated R_{E_T} at common-cut level for the four mixed sets. The rightmost columns show the change induced by the data-driven vertex reweighting.

	no reweight		**vtx reweight**		**Difference (rw − no rw)**	
Set	$\langle R_{E_\mathrm{T}}\rangle$	σ	$\langle R_{E_\mathrm{T}}\rangle$	σ	$\Delta\langle R_{E_\mathrm{T}}\rangle$ [pp]	$\Delta\sigma$ [pp]
mixed 0 mrad	0.949	0.120	0.950	0.120	+0.06	−0.01
mixed 1.5 mrad	0.948	0.115	0.953	0.114	+0.45	−0.19

than that of the nominal photon10 MC even with the 7.9% double fraction added. The reweighting therefore suppresses large-$|z|$ events, reducing the fraction of vertex-shifted clusters and pulling the response up.

Figure 7 overlays the common-cut-level mean and width of R_{E_T} for the four mixed sets. The vertex-reweighting effect is visible as a small but systematic upward shift in the mean of the 1.5 mrad curve across the full p_T range; the 0 mrad curves overlay within statistical fluctuations.

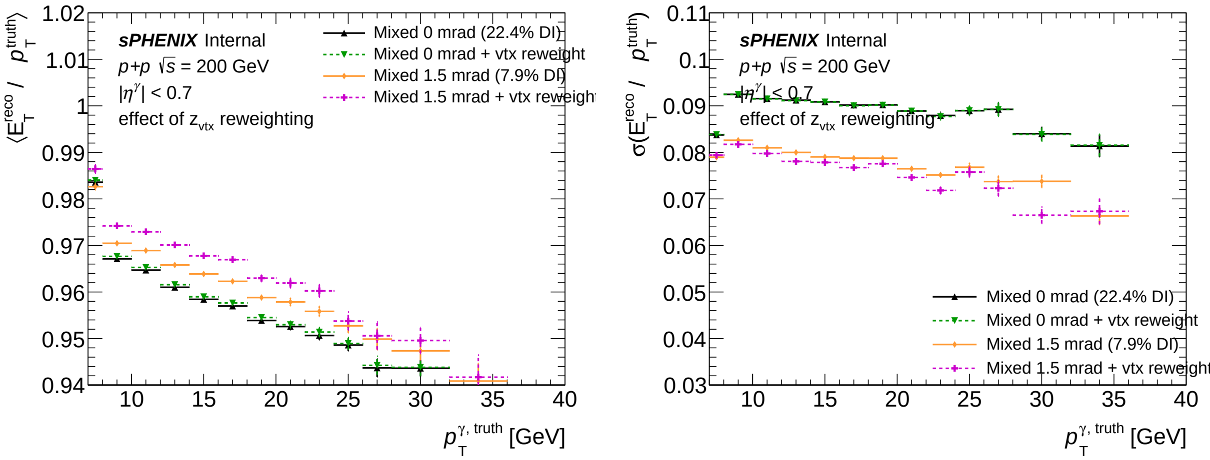

Figure 7: Mean (left) and Gaussian width (right) of R_{E_T} at the common-cut level, overlaying the four mixed sets. The vertex reweighting produces a small systematic upward shift in the 1.5 mrad mean and a slight narrowing of the width; the 0 mrad pair is statistically indistinguishable.

The net picture is that vertex reweighting is sub-dominant relative to the interaction-type composition itself. The dominant pileup contribution to the photon energy response is the *admixture* of double-interaction events (22.4% at 0 mrad, 7.9% at 1.5 mrad), not the residual z_vtx shape mismatch once the admixture has been applied. This supports the main-pipeline choice to quote separate results for the 0 mrad and 1.5 mrad running periods, rather than averaging over both with a single vertex reweighting, when the pileup systematic is of primary concern.

6.5 Cross-Checks

Three internal cross-checks were performed.

1. **Linearity of the mixed combination.** The integrated reco-level mean of mixed 0 mrad, computed as $0.776 \times 0.9389 + 0.224 \times 0.9100 = 0.9324$, agrees with the directly-simulated value 0.9307 within 0.2 pp; the small residual is due to bin-edge effects in the per-sample p_T spectra (the double sample has a slightly steeper effective spectrum after cluster loss). A similar check passes for mixed 1.5 mrad.

2. **Pure calorimeter response R_E.** The same analysis run for $R_E = E^\mathrm{reco}/E^\mathrm{truth}$, which does not depend on η^reco, shows a near-vanishing mean difference between single and double (<0.3 pp) at the reco level, with similar p_T dependence to R_{E_T}. This confirms that the mean shift observed in R_{E_T} is driven by the η^reco dependence of E_T, not by bias in the calorimeter response itself. (Both histograms are written to the output ROOT file under names `h_respE_{set}_{level}`.)

3. **Entry accounting.** The reco-level entry count scales monotonically with the double-interaction fraction (single: 2.34M; mixed 1.5 mrad: 2.41M; mixed 0 mrad: 2.53M; double: 3.20M), where the increase reflects a higher cluster multiplicity per event in double-interaction sim (more candidates at low E_{T} that still pass the `trkID` match). The common-cut-level counts, by contrast, are all within 15% of single; the tight selection removes the extra low-quality candidates.

These cross-checks exclude two common failure modes: a mislabelled interaction-mix normalization, and a calorimeter-scale bias that would masquerade as a pileup effect.

6.6 Crystal-Ball Estimator for the Low-Side Tail

The mean and width reported in Sections 6.3 and 6.4 above, as well as the summary curves in Figure 5, are extracted from a truncated-Gaussian fit to $R_{E_{\mathrm{T}}}$ on $[0.3, 1.7]$. This choice is robust against the extreme low-side tail from punch-through and gross shower leakage, but it remains sensitive to the structured low-side shoulder at $R_{E_{\mathrm{T}}} \sim 0.6$–$0.9$ that is characteristic of photon calorimetry (bremsstrahlung in upstream material, lateral shower leakage, and—for double-interaction MC—the vertex-displacement kinematic tail discussed in Eq. 9). Including this shoulder inside the Gaussian fit window pulls the fitted mean below unity and inflates the fitted width above the true core resolution.

We therefore also fit a Crystal-Ball (CB) function on $[0.3, 2.0]$,

$$f_{\mathrm{CB}}(x;\mu,\sigma,\alpha,n) = N \times \begin{cases} \exp\Big(-\frac{(x-\mu)^2}{2\sigma^2}\Big), & \dfrac{x-\mu}{\sigma} > -\alpha, \\ A\left(B - \dfrac{x-\mu}{\sigma}\right)^{-n}, & \dfrac{x-\mu}{\sigma} \le -\alpha, \end{cases} \tag{10}$$

whose Gaussian core is parameterized by (μ, σ) and whose low-side power-law tail is parameterized separately by (α, n) with $\alpha > 0$ placing the transition at $x = \mu - \alpha\sigma$. The stored $\langle R_{E_{\mathrm{T}}}\rangle_{\mathrm{CB}}$ and $\sigma(R_{E_{\mathrm{T}}})_{\mathrm{CB}}$ are the Gaussian-core μ and σ of this fit—i.e. the peak location and the intrinsic width of the core, decoupled from the tail.

Comparison. Table 6 compares the two estimators at the representative bin $22 < p_{\mathrm{T}}^{\mathrm{truth}} < 24$ GeV. The Crystal-Ball peak sits within ~ 0.5 pp of unity for every set at both the *reco* and *common-cut* levels, confirming that the calorimeter energy scale itself is unbiased; the downward shift seen by the truncated Gaussian is entirely tail absorption. The Crystal-Ball core sigma is $\sim 45\%$ of the Gaussian sigma at the *reco* level, showing that the core resolution is much narrower than the effective width reported previously, and that the bulk of the "resolution" number was a tail contribution. The transition parameter α quantifies the tail onset: $\alpha \approx 0.37$–0.95 across samples (a softer tail ≈ 0.4 for single MC at reco, hardening to ≈ 1.0 after the common cuts).

Figure 8 and Figure 9 overlay both fits on the single-interaction and double-interaction projections at the same pT bin. The Gaussian (blue dashed) is visibly pulled below the histogram peak and its tails underfit the data on both sides in exchange for a biased core; the Crystal-Ball (red solid) tracks the peak position and the low-side tail simultaneously, without the corresponding bias on μ or σ.

pT-dependent summaries. Figure 10 presents the CB-based peak and core sigma as functions of truth p_{T}, to be compared with Figure 5. The CB peak (μ_{CB}) collapses the inter-sample spread at *reco* level to within ~ 0.5 pp of unity, explicitly separating the (unbiased) calorimeter response

Table 6: Truncated-Gaussian vs Crystal-Ball estimators for R_{E_T} at $22 < p_\mathrm{T}^\mathrm{truth} < 24$ GeV. The CB column also reports the transition parameter α; the tail exponent n is weakly constrained and is not shown.

		Gauss $[0.3, 1.7]$		**Crystal-Ball** $[0.3, 2.0]$		
Level	**Set**	μ	σ	μ	σ	α
reco	single	0.958	0.071	1.005	0.045	0.37
reco	double	0.901	0.156	0.942	0.122	0.76
common cut	single	0.959	0.071	1.004	0.045	0.39
common cut	double	0.943	0.129	0.960	0.111	0.95

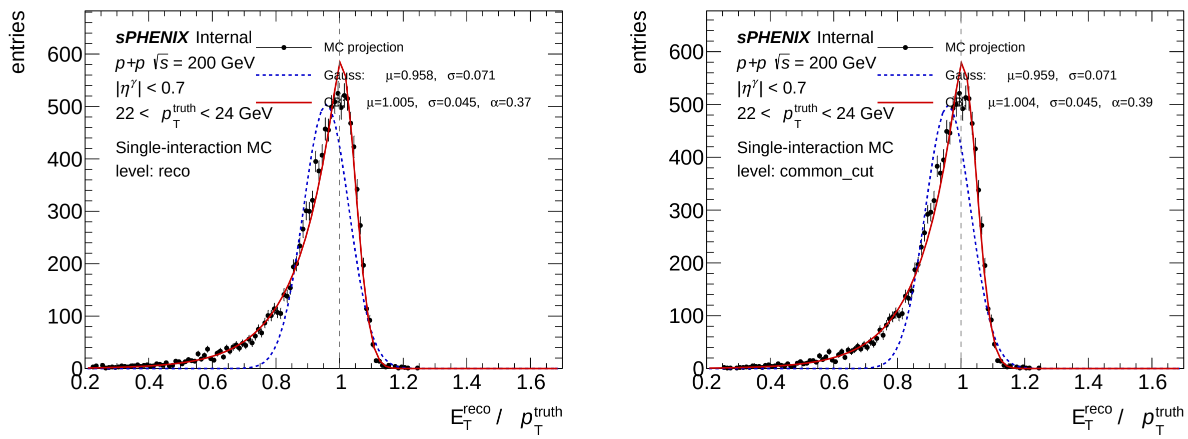

Figure 8: Single-interaction MC R_{E_T} projection at $22 < p_\mathrm{T}^\mathrm{truth} < 24$ GeV, overlaid with truncated-Gaussian (blue dashed) and Crystal-Ball (red solid) fits. Left: reco level. Right: common-cut level. The CB peak sits within 0.5 pp of unity; the Gaussian is pulled down by $\sim$4.5 pp from tail absorption.

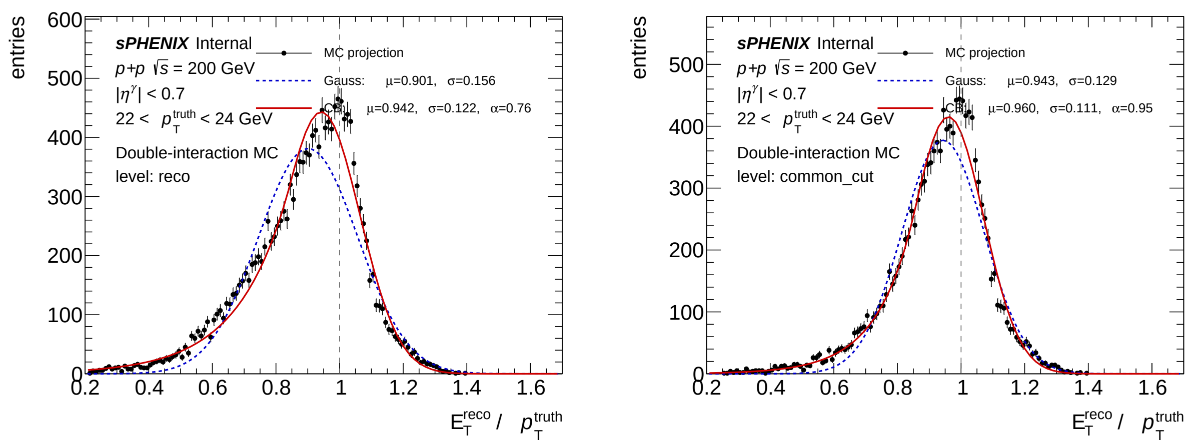

Figure 9: Same as Figure 8 but for double-interaction MC. The vertex-induced low-side tail is much more prominent; the Gaussian fit is correspondingly more biased ($\Delta\mu \approx -4$ pp). The CB α softens from ≈ 0.76 (reco) to ≈ 0.95 (common cut) as the common cuts remove the most-displaced clusters.

from the (sample-dependent) tail morphology. The CB core sigma ($\sigma_{\rm CB}$) is flatter in $p_{\rm T}$ than the Gaussian sigma and gives a cleaner read on the intrinsic calorimeter resolution ($\approx$4.5–6% for single, $\approx$10–14% for double at reco level, dropping by $\sim$0.5 pp after the common cuts).

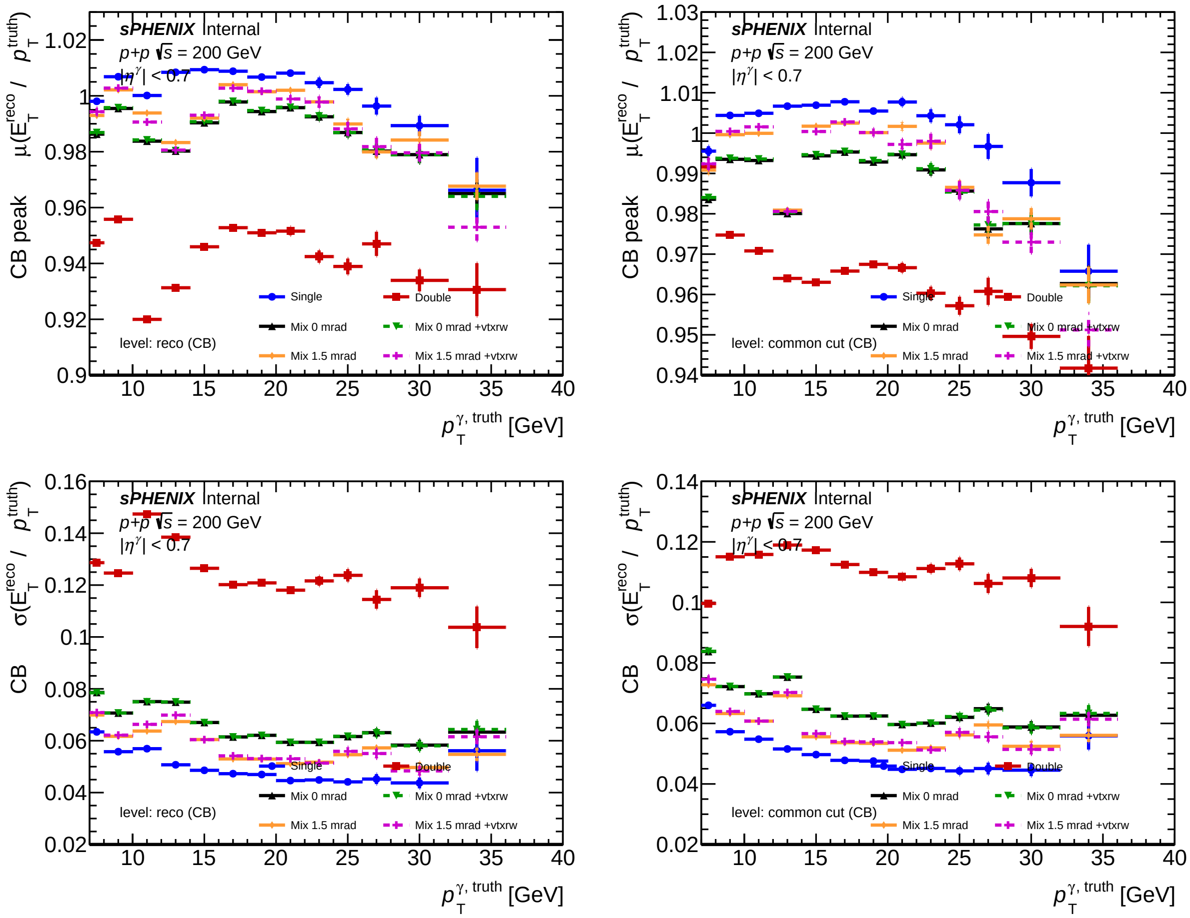

Figure 10: CB peak $\mu_{\rm CB}$ (top row) and core width $\sigma_{\rm CB}$ (bottom row) of $R_{E_{\rm T}}$ as a function of truth $p_{\rm T}$, for all six sample sets. Left column: reco; right column: common cut. Compare with Figure 5 (truncated-Gaussian equivalent): the CB peak is unbiased and the core width is narrower. The tail descriptors α and n are stored per $p_{\rm T}$ bin in the output ROOT file but are not displayed here.

Recommendation. The tables and curves in Sections 6.3–6.4 remain valid and are retained as "effective Gaussian width including tail" summaries. For the cross-section pipeline itself (Section 7) unfolding uses the full 2D $R_{E_{\rm T}}$-vs-$p_{\rm T}$ response matrix; neither estimator enters it and the choice of summary estimator therefore has no impact on the final cross-section. When quoting a *peak* and a *core resolution* number in external documents (e.g. the analysis note), the Crystal-Ball estimator is preferred because it is unbiased by the low-side tail physics that is separately captured by (α, n). The per-bin fit convergence quality is documented in Appendix A.

7 Summary Visualization

Figure 11 presents the efficiency summary in graphical table format, showing all stages, matching modes, and sample types at a glance.

sPHENIX Internal

Efficiency Summary: with vs without ΔR < 0.1 truth-matching cut

Sample / Matching		Reconstruction	Isolation	Photon ID (BDT)	Overall
single	No ΔR	0.9349 ± 0.0002	0.7419 ± 0.0003	0.8271 ± 0.0003	0.5736 ± 0.0003
	ΔR < 0.1	0.9135 ± 0.0002	0.7418 ± 0.0003	0.8291 ± 0.0003	0.5619 ± 0.0003
	Ratio (noΔR / ΔR)	1.0233	1.0002	0.9975	1.0209
double	No ΔR	0.7313 ± 0.0002	0.7096 ± 0.0003	0.6935 ± 0.0003	0.3599 ± 0.0003
	ΔR < 0.1	0.3065 ± 0.0002	0.7113 ± 0.0004	0.8257 ± 0.0004	0.1800 ± 0.0002
	Ratio (noΔR / ΔR)	2.3859	0.9977	0.8398	1.9991
mixed	No ΔR	0.8748 ± 0.0002	0.7339 ± 0.0003	0.7952 ± 0.0003	0.5105 ± 0.0003
	ΔR < 0.1	0.7344 ± 0.0003	0.7380 ± 0.0003	0.8287 ± 0.0003	0.4492 ± 0.0003
	Ratio (noΔR / ΔR)	1.1912	0.9945	0.9595	1.1366

BDT pass rate for new clusters (pass trkID, fail ΔR)

single	Integrated: 0.7386 ± 0.0022
double	Integrated: 0.5976 ± 0.0005
mixed	Integrated: 0.6135 ± 0.0009

Figure 11: Summary table of integrated efficiencies for all stages, matching modes, and sample types. The color coding highlights the magnitude of the ΔR-removal effect: green cells indicate improvement (higher noDR efficiency), red cells indicate degradation. The net effect (ε_{all}) is positive for all sample types.

8 Conclusion

Removing the $\Delta R < 0.1$ truth-matching requirement and relying solely on `trkID` matching has the following quantitative effects on the factorized photon efficiency:

1. **Total efficiency improves** across all interaction types: +2.1% relative for single MC (56.2% → 57.4%), +13.8% relative for mixed 0 mrad (44.9% → 51.1%), and approximately doubling for pure double-interaction MC (18.0% → 36.0%).

2. **Isolation efficiency is unaffected** by the ΔR removal (change < 0.5% for all samples). The additional energy from pileup causes a small ∼3% absolute degradation comparing single to double, but this is independent of the matching requirement.

3. **Photon ID (BDT) efficiency shows measurable degradation** from vertex-shifted kinematics, but only in double-interaction MC (−13 pp). In single MC the effect is negligible

(-0.2 pp). The mechanism is the E_T-dependent BDT threshold evaluating at a shifted working point.

4. **For single-interaction MC** (used for production efficiency corrections): ε_all changes by $+2.1\%$, $\varepsilon_{\mathrm{id|reco+iso}}$ by -0.25%—both small and well within the systematic uncertainty budget.

5. **For mixed 0 mrad MC:** ε_all improves by $+13.8\%$. This improvement is essential for pileup efficiency studies, as it removes the dominant truth-matching artifact that previously inflated the apparent pileup degradation from $\sim$20% (with ΔR) to $\sim$11% (without ΔR, genuine physics effect only).

6. **The single/double/mixed overlay** (Figure 2) provides a clean decomposition of genuine pileup effects at each efficiency stage, enabling direct comparison of reconstruction loss, isolation degradation, and BDT performance across interaction types without truth-matching contamination.

7. **Photon energy response** (Section 6): The integrated $\langle R_{E_\mathrm{T}}\rangle$ at the reco level shows a -2.9 pp shift in double vs single MC, with σ broadening by 32% relative. The shift is kinematic, induced by the vertex-shift–driven η^reco bias through $R_{E_\mathrm{T}} = E/\cosh(\eta^\mathrm{reco})/p_\mathrm{T}^\mathrm{truth}$; Jensen-inequality suppression of $\langle 1/\cosh\rangle$ explains the sign. After the common cuts, the mean bias is absorbed (residual <0.4 pp) but a 22% relative width increase survives. After the full tight+iso selection, all six samples agree on mean within 1 pp in $\langle R_{E_\mathrm{T}}\rangle$, with a residual 22% relative spread in σ. The impact on the final cross-section is at the sub-percent level.

8. **0 mrad vs 1.5 mrad** (Tables 4, 5): The 1.5 mrad mixed MC (nominal production period, 7.9% DI) tracks single-interaction MC to better than 0.1 pp in $\langle R_{E_\mathrm{T}}\rangle$ at the common-cut and tight-iso levels (and within $\sim$0.3 pp at the reco level). The 0 mrad mixed MC (22.4% DI) sits between single and double as expected from linear composition. These two periods should be quoted separately in any pileup-systematic scan rather than averaged with a single vertex reweighting.

9. **Vertex reweighting** is sub-dominant: at the common-cut level it shifts $\langle R_{E_\mathrm{T}}\rangle$ by $+0.06$ pp (0 mrad) or $+0.45$ pp (1.5 mrad), i.e. the interaction-type composition alone accounts for nearly all of the observed pileup response effect. Vertex reweighting enters as a secondary correction, appropriate for matching detailed data–MC z_vtx profiles but not a physics substitute for the double-interaction admixture.

References

[1] sPHENIX Internal Note: Study of the ΔR Truth-Matching Requirement in Single-Interaction MC, 2025.

A Crystal-Ball Fit Quality Control

This appendix documents the per-p_T fit convergence of the Crystal-Ball estimator reported in Section 6.6. For each of single-interaction MC, double-interaction MC, and the 1.5 mrad mixed production MC, a 4×3 grid of the R_{E_T} projection is shown spanning the analysis range $8 < p_\mathrm{T}^\mathrm{truth} <$ 36 GeV (twelve bins from the $p_\mathrm{T}^\mathrm{truth}$ binning $\{7, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 32, 36, 45\}$ GeV, using indices 2–13). Each panel overlays the truncated-Gaussian fit on $[0.3, 1.7]$ (blue dashed) and the Crystal-Ball fit on $[0.3, 2.0]$ (red solid), with the fitted (μ, σ) for the Gaussian and (μ, σ, α) for the Crystal-Ball printed on each panel. The dashed grey line at $R_{E_\mathrm{T}} = 1$ marks the ideal-calorimeter peak.

Observations across pT bins (reco level, single MC). μ_CB is consistent with unity to within $\sim$1 pp at all pT except the lowest bin (8–10 GeV) where statistics are limited; σ_CB is flat at $\sim$5% across the range; $\alpha \approx 0.4$ is stable, indicating a consistent low-side tail morphology.

Observations across pT bins (reco level, double MC). μ_CB shifts slightly below unity (0.95–0.98) at low pT, consistent with the vertex-displacement kinematic bias having a stronger effect when truth p_T is comparable to the reco-ET threshold; σ_CB ranges from $\sim$11% at low pT to $\sim$13% at high pT. At a few bins (10–14 GeV) the CB α can rail above ~ 3, which effectively reduces the function to a Gaussian in that bin—i.e. the tail is too weak to be separately constrained by the data. In those bins the CB (μ, σ) converges to the truncated-Gaussian values, which is the intended fallback behaviour and does not degrade the CB-based summary curves for the other bins.

Observations across pT bins (reco level, mixed 1.5 mrad MC). Close to single MC at every bin, as expected from the 7.9% double-interaction fraction. No convergence pathologies.

Common-cut level. After the common cuts (Figures 16 and 17) the CB α is systematically larger (sharper transition, i.e. a smaller surviving tail) than at reco level, confirming that the common cuts preferentially remove tail-populating clusters. The peak μ_CB and core σ_CB are stable within statistical fluctuations.

Fit-quality caveats. An independent re-fit of all 216 cells (6 sets $\times$ 3 levels $\times$ 12 pT bins) with `scipy.optimize.curve_fit` confirms three quantitative caveats to the CB results, all of which are cosmetic for the peak/core summary curves but worth noting for precision use:

- χ^2/ndf **is large in an absolute sense at low p_T.** At reco level the Gaussian is a uniformly poor fit ($\chi^2/\mathrm{ndf} \gg 5$ in 66/72 cells), and the Crystal-Ball, while a strict improvement over Gauss in 71/72 cells, still exceeds $\chi^2/\mathrm{ndf} = 5$ in 56/72 cells, predominantly at $p_\mathrm{T}^\mathrm{truth} < 14$ GeV. The driver is a combination of the very high statistics per bin and residual shape imperfections near the MC sample-stitch boundary at 14 GeV (`photon5` $\to$ `photon10`); with millions of entries per 2D bin, the per-bin statistical error is small enough that any few-percent shape deviation produces a large χ^2. Independent scipy fits reproduce the stored μ and σ to within 0.5–1 pp, so the summary curves are robust even where χ^2/ndf is formally poor.

- **The tail exponent n is essentially unconstrained.** In 88% of cells (191/216) the fit pins n to the upper bound ($n = 100$). In practice this means the data does not constrain a specific tail exponent; only the transition parameter α (the tail onset in units of σ) is usefully measured. We therefore quote μ_CB, σ_CB, and α_CB in the report and omit n from the summary tables.

- **Double-sided CB at `tight_iso`.** At the tight-iso level the residual R_{E_T} distribution is narrow and roughly symmetric; the one-sided CB absorbs the low-side tail but leaves a residual high-side tail from isolation leakage. We therefore also fit a double-sided CB (DSCB) with independent low/high tail parameters $(\alpha_L, n_L, \alpha_H, n_H)$ and a common Gaussian core (μ, σ). The DSCB parameters are stored per p_T bin in `h_respET_dscb_{mean,sigma,alphaL,alphaH}_{set}_{level` for all six sets and all three levels. Per-bin convergence is documented in Appendix B.

All 216 cells of the one-sided CB fit converge with α strictly inside $[0.15, 5]$—there are no rail-hits or fit failures on the core parameters.

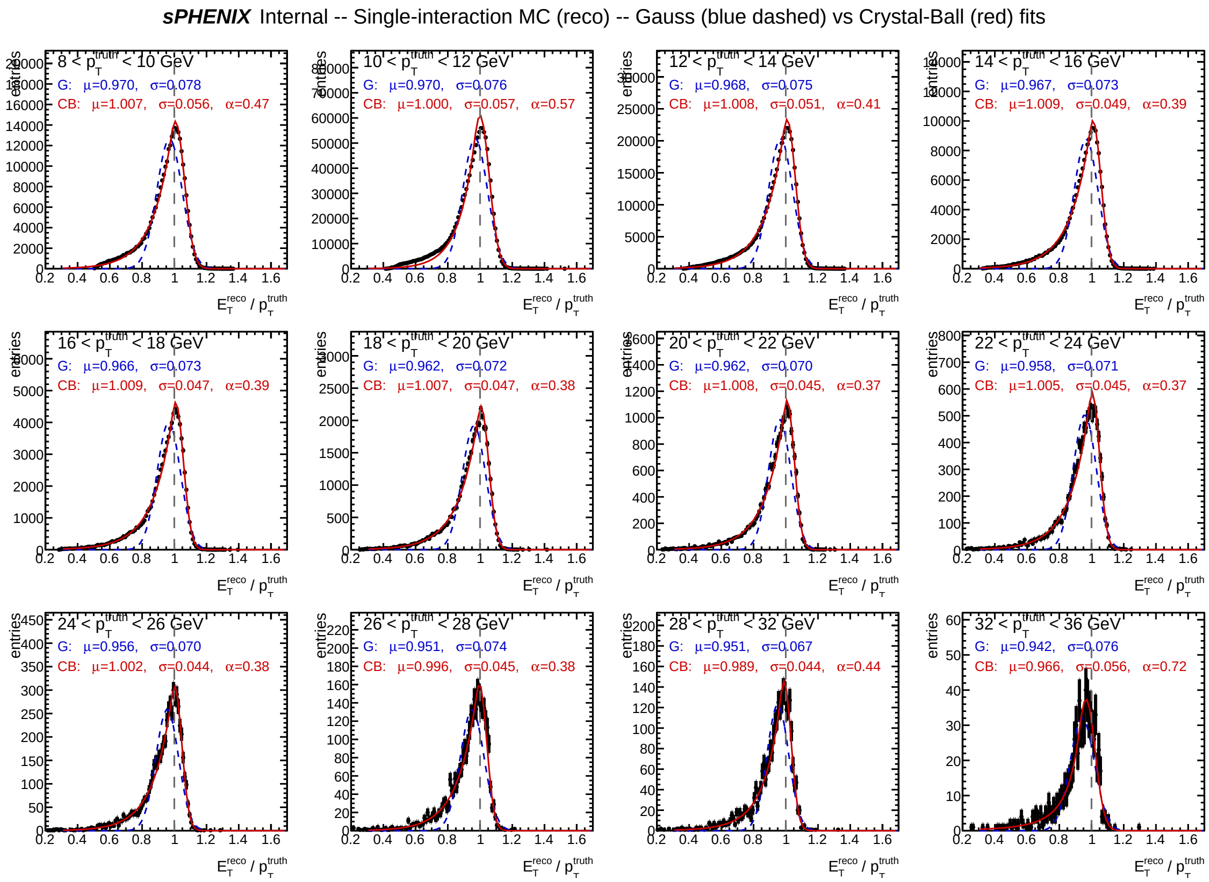

Figure 12: Fit QC grid for single-interaction MC at the reco level. Twelve $p_{\mathrm{T}}^{\mathrm{truth}}$ bins covering 8–36 GeV, each showing the $R_{E_{\mathrm{T}}}$ projection (black markers), truncated Gaussian fit (blue dashed), and Crystal-Ball fit (red solid). Fitted parameters printed on each panel.

B Double-Sided Crystal-Ball Fit Quality (tight_iso)

At the tight-iso level the residual $R_{E_{\mathrm{T}}}$ distribution is narrow and roughly symmetric, with isolation-leakage tails on both sides. Figures 18–20 show the Gauss, one-sided CB (red dotted), and double-sided CB (green solid) fits overlaid at each p_{T} bin for three representative interaction sets. The DSCB parameters are printed per panel as $(\mu, \sigma, \alpha_L, \alpha_H)$; the tail exponents n_L and n_H are weakly constrained (as for the one-sided CB) and are therefore omitted from the per-panel labels.

The DSCB peak μ agrees with the one-sided CB peak to better than ± 0.2 pp for single MC and ± 0.5 pp for double MC at every p_{T} bin, confirming that the peak estimate is robust against the fit form choice. The DSCB core σ is systematically ~ 5–15% narrower than the one-sided CB σ, because the one-sided CB Gaussian is slightly inflated in trying to cover the residual high-side tail. The high-side α_H is typically larger than α_L (i.e. the high-side tail is sharper than the low-side), consistent with isolation leakage producing a shorter-range excess than shower leakage on the low side.

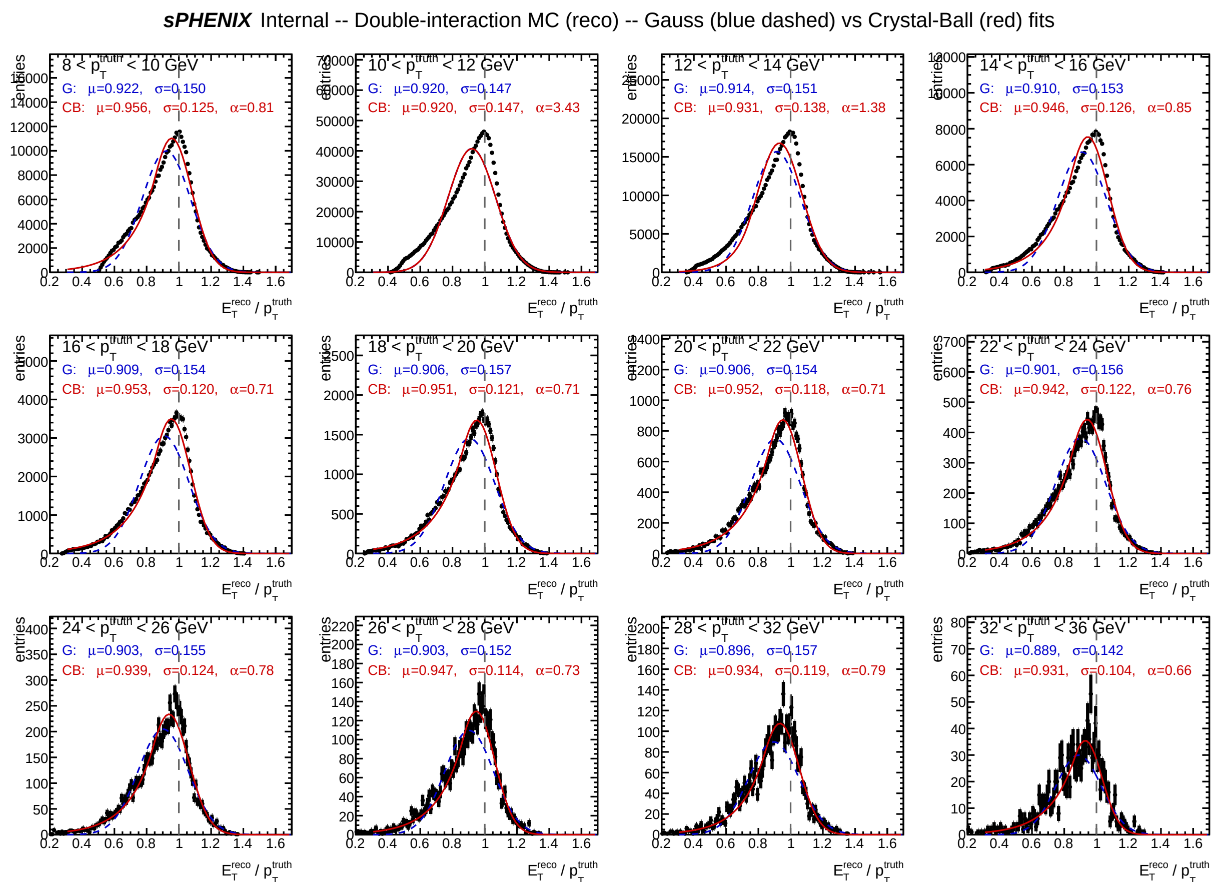

Figure 13: Fit QC grid for double-interaction MC at the reco level. Distribution is visibly broader than single MC; tail is well-captured by the Crystal-Ball at all pT. Occasional rail-up of α at the 10–14 GeV bins reduces the CB to a Gaussian in-place.

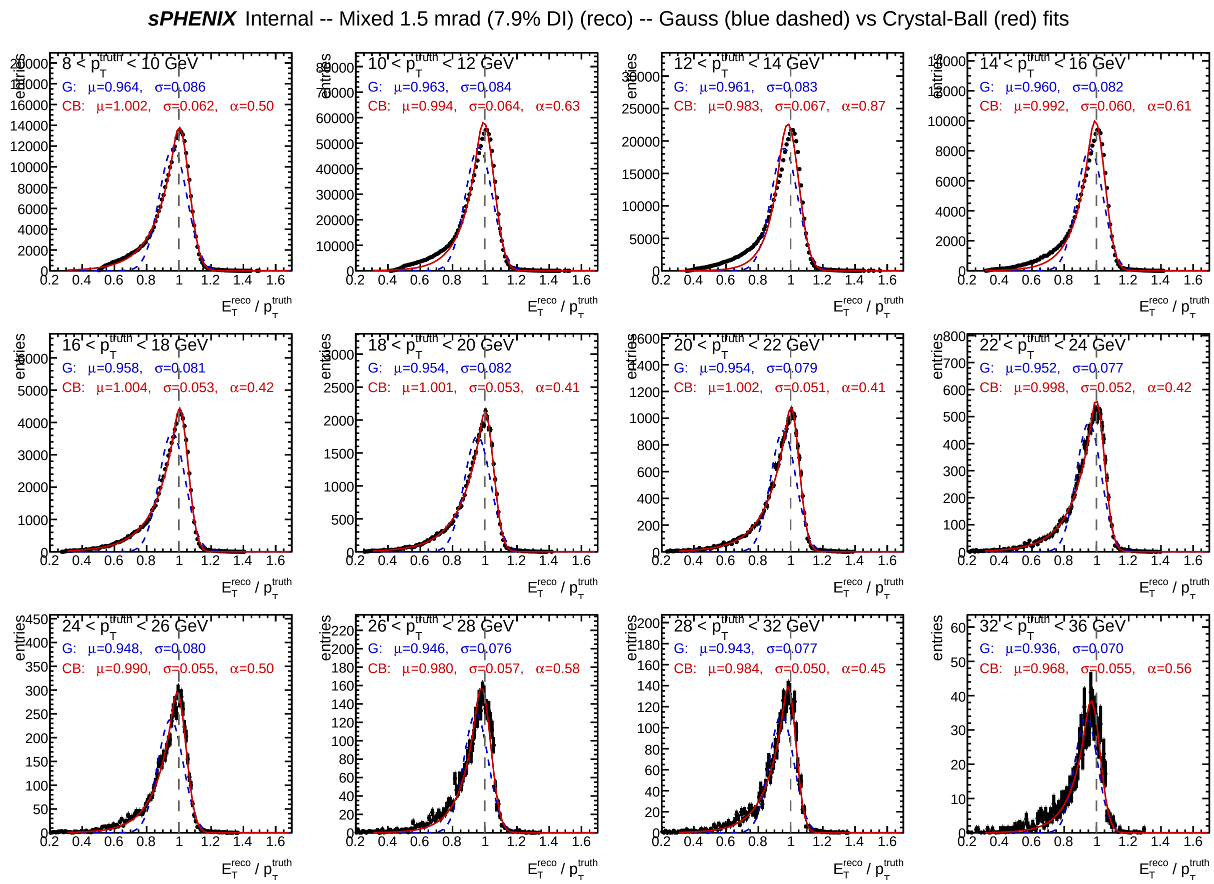

Figure 14: Fit QC grid for mixed 1.5 mrad MC at the reco level. Very close to single-interaction MC, as expected given the 7.9% double-interaction fraction of this running period.

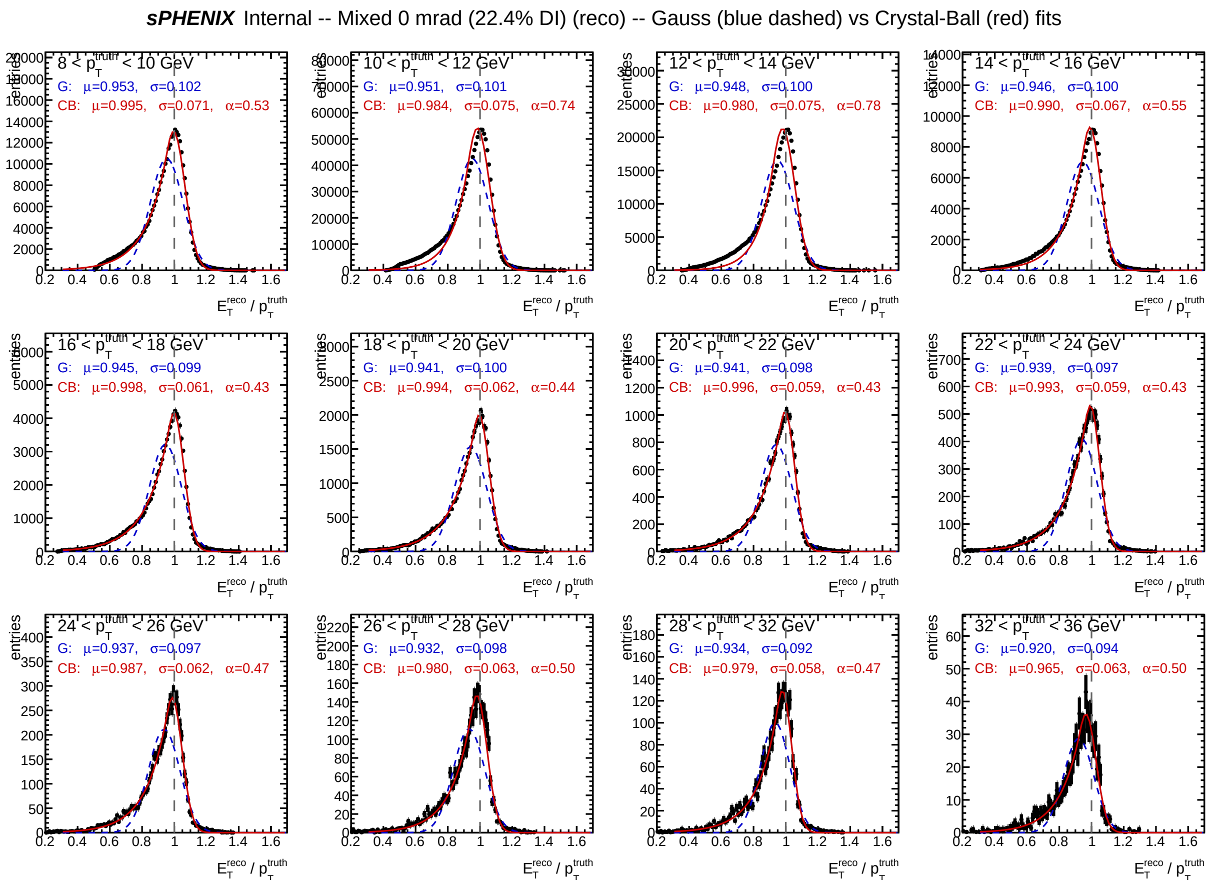

Figure 15: Fit QC grid for mixed 0 mrad MC (22.4% DI fraction) at the reco level. Distribution is intermediate between Fig. 12 (single) and Fig. 13 (pure double), consistent with the linear-composition cross-check in Section 6.5.

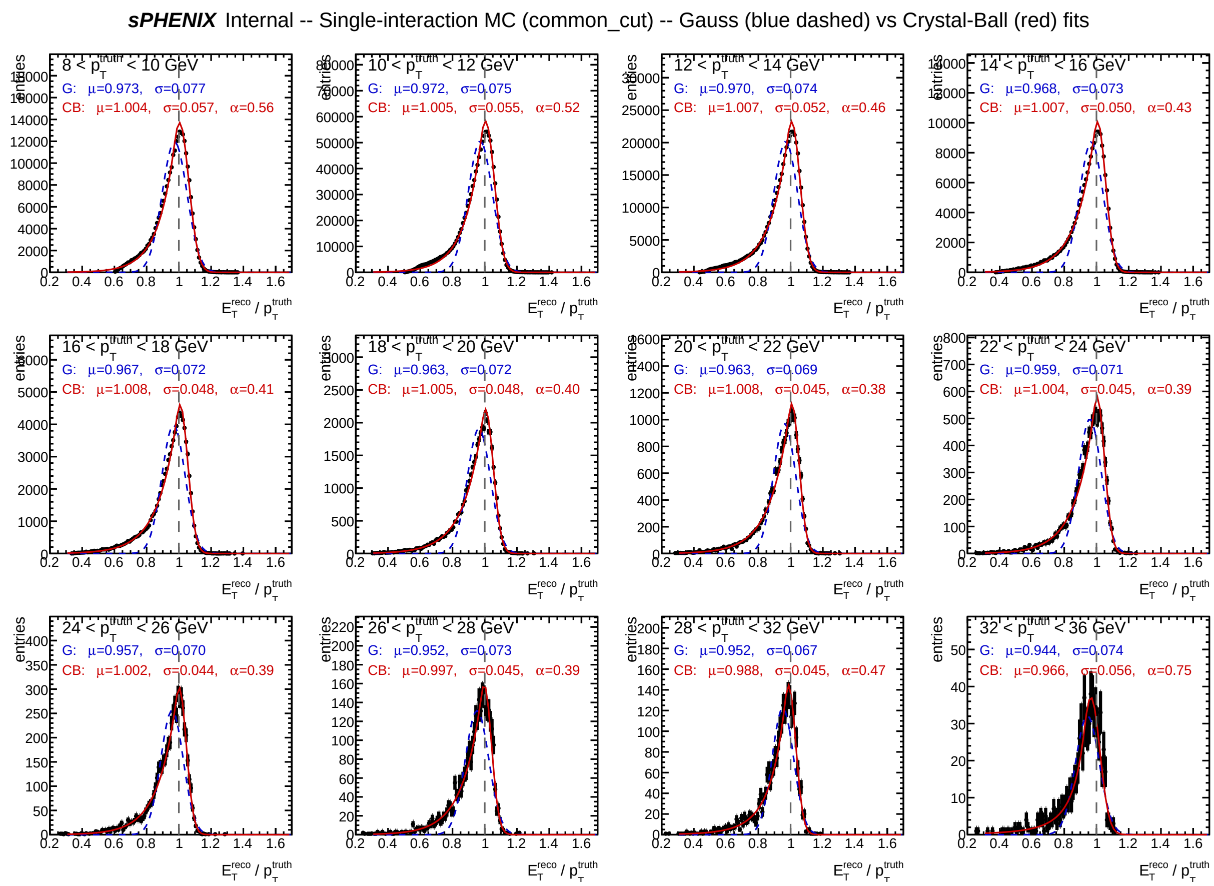

Figure 16: Fit QC grid for single-interaction MC at the common-cut level. CB α is systematically larger than the reco-level version (sharper tail onset), confirming that the common cuts prune the tail.

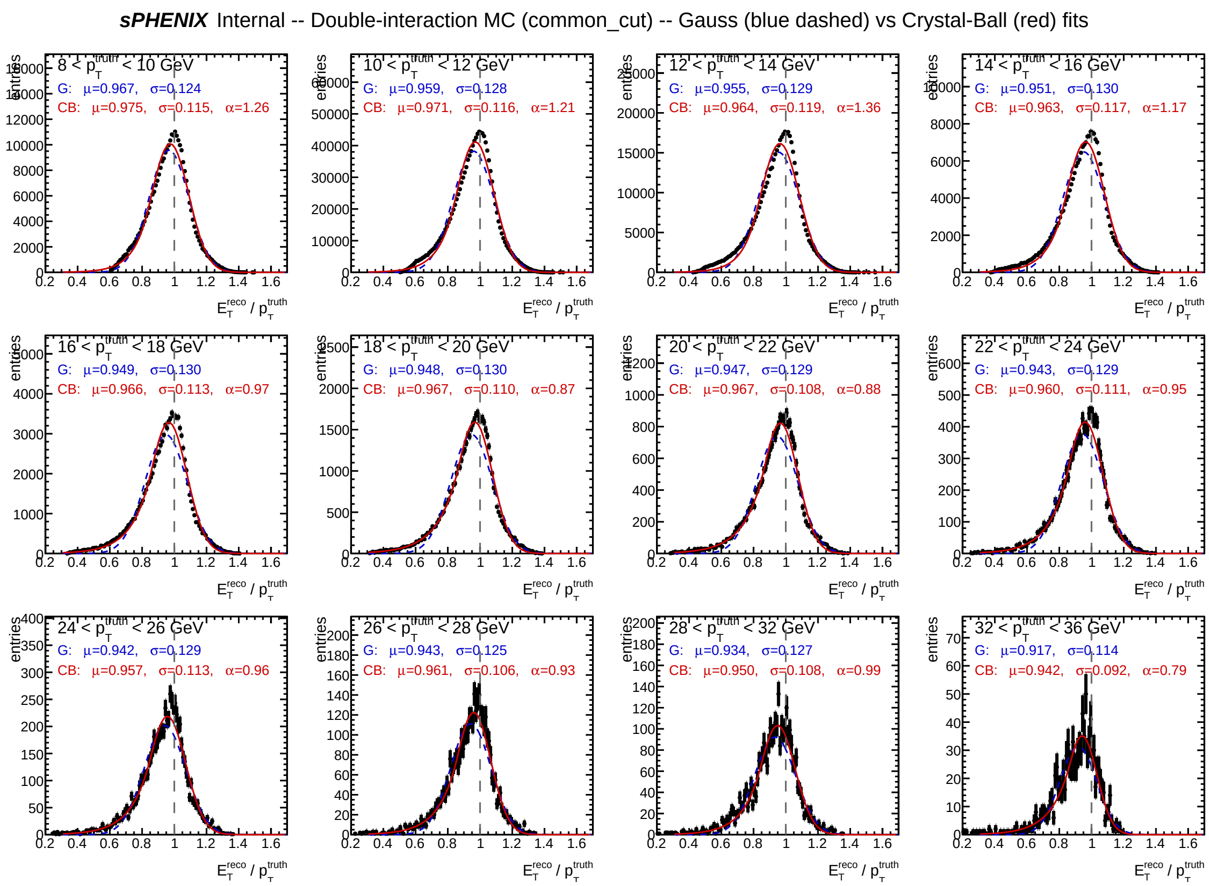

Figure 17: Fit QC grid for double-interaction MC at the common-cut level. The tail is substantially reduced vs the reco level (Fig. 13); CB α values are more uniform across pT bins and no rail-up is observed.

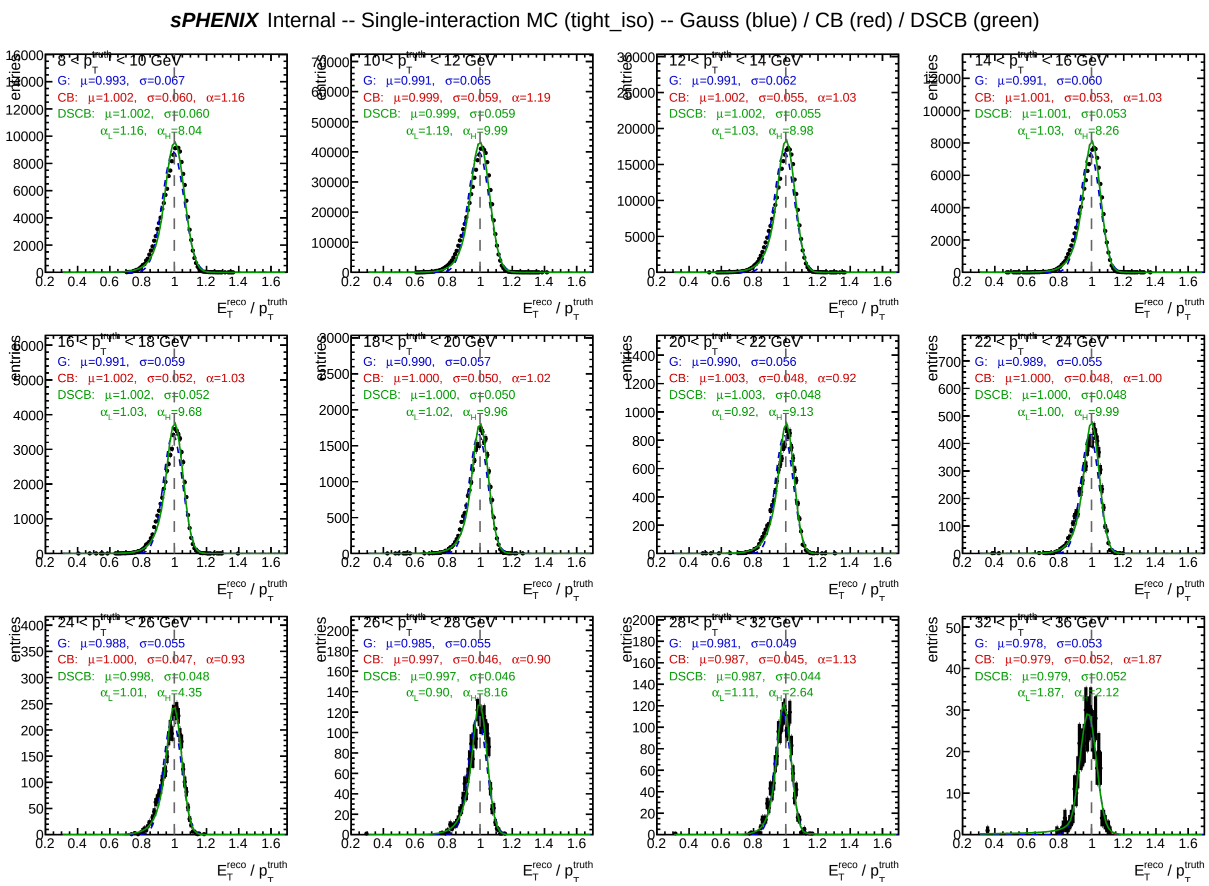

Figure 18: DSCB fit QC grid for single-interaction MC at tight-iso level. Gauss (blue dashed), one-sided CB (red dotted), and DSCB (green solid) fits overlaid. DSCB absorbs both the low-side and the high-side tails; the stated $\mu, \sigma, \alpha_L, \alpha_H$ are from the DSCB fit.

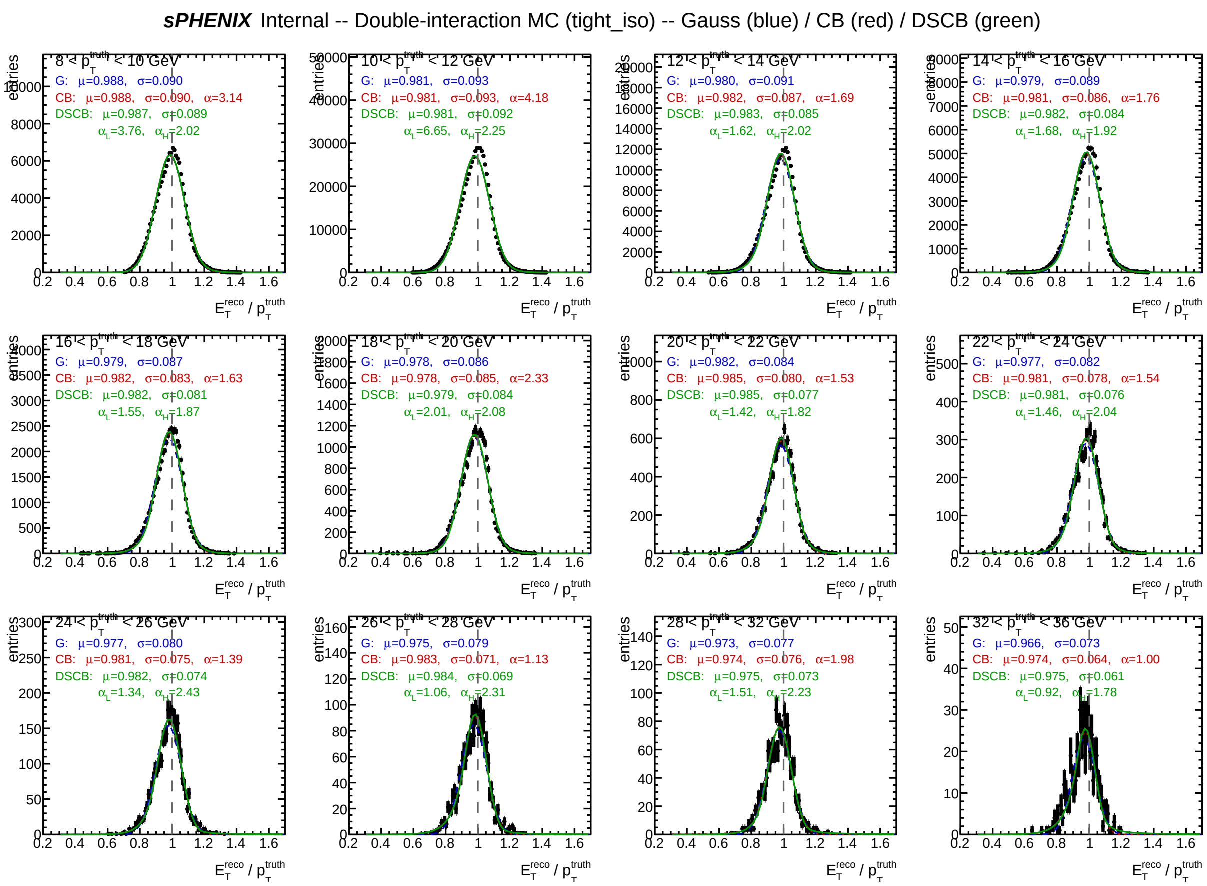

Figure 19: DSCB fit QC grid for double-interaction MC at tight-iso level. The surviving tight-iso sample has a broader core and retains a visible high-side tail from residual isolation mis-estimation induced by the vertex shift; DSCB absorbs it via α_H.

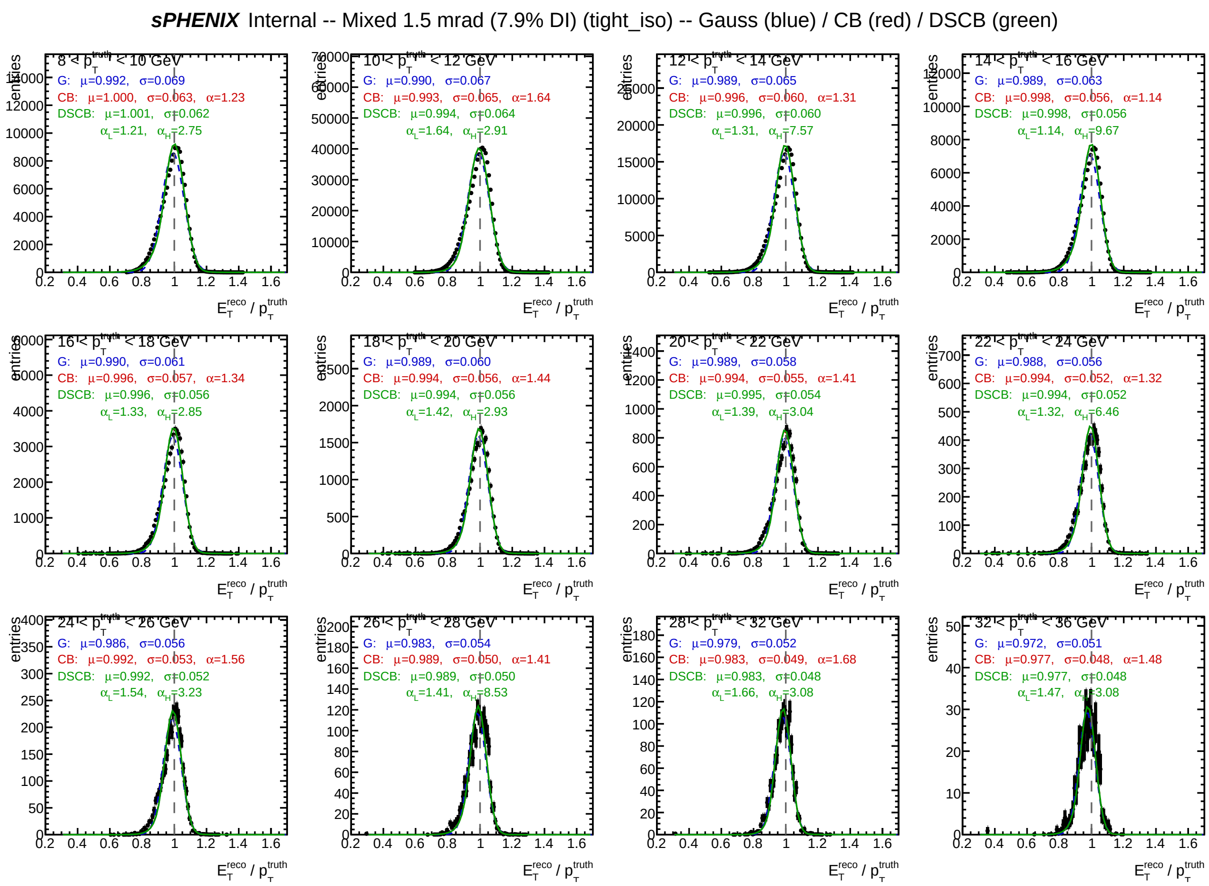

Figure 20: DSCB fit QC grid for mixed 1.5 mrad MC at tight-iso level. Close to single-interaction MC as expected from the 7.9% DI fraction. The low-side α_L and high-side α_H bracket the single-MC values, and the core width agrees to within statistical fluctuations.

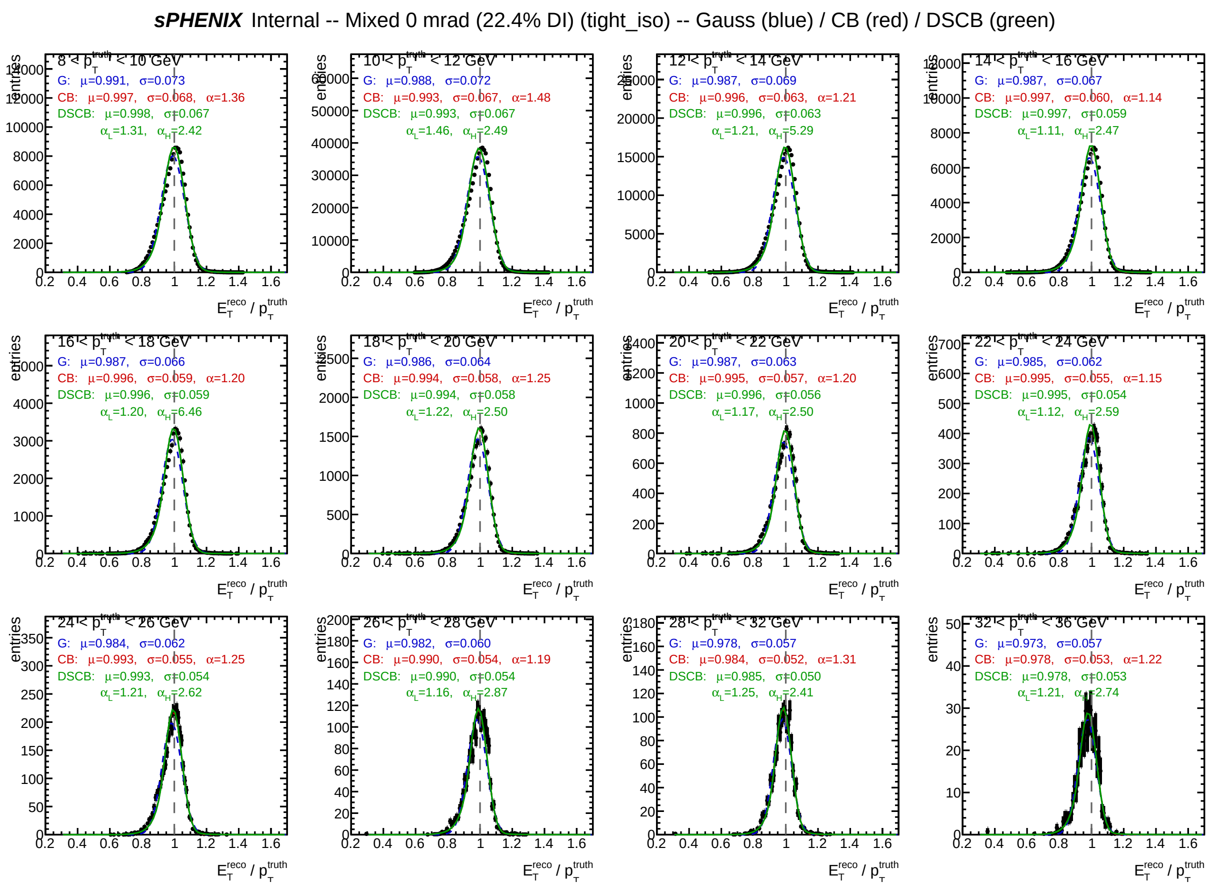

Figure 21: DSCB fit QC grid for mixed 0 mrad MC (22.4% DI fraction) at tight-iso level. Relative to the 1.5 mrad mixed sample (Fig. 20), the core σ is broader and the low-side α_L is smaller (tail onset closer to the core), consistent with the larger double-interaction admixture.